

School Construction and Dimensions of Adult Depressive Symptoms: Evidence from Indonesia's INPRES Program

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Abstract

I examine whether exposure to Indonesia's INPRES school construction program is associated with depressive symptoms in adulthood decades later. Using pooled Indonesian Family Life Survey data from 13 provinces (8,821 person-wave observations from 5,144 individuals), I relate district-level program intensity to CES-D depression scores. Depressive symptoms are lower among women from higher-intensity districts, while estimates for men do not show a comparable pattern. An item-level analysis shows that the association is concentrated in fearfulness and "could not get going," rather than in positive-affect items. Because the program bundled school construction with other local investments and produced little detectable change in female schooling in this sample, the estimates are best interpreted as long-run associations of program exposure rather than returns to schooling.

Keywords: School construction, Depression, Reduced-form, CES-D, Indonesia, INPRES

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Highlights

- Exposure to higher-intensity INPRES districts is associated with lower female CES-D scores decades later

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- The female reduced-form result is best supported because male estimates fail pre-trends checks
- Item-level associations are largest for fearfulness and “could not get going” rather than positive affect
- Near-zero female first stage is inconsistent with detectable schooling gains as the main channel in this sample
- Findings are consistent with bundled community-level channels but do not isolate schooling itself

1. INTRODUCTION

Is exposure to Indonesia’s INPRES school construction program associated with lower depressive symptoms decades later? That question matters because depressive symptoms can affect labor supply, productivity, and household welfare, yet quasi-experimental evidence on the long-run mental-health associations of school expansion remains limited. The causal literature on education and depression is mixed, and most of it comes from compulsory schooling reforms in high-income settings. European reforms have produced both protective (Crespo et al., 2014; Chevalier and Feinstein, 2006) and adverse associations (Avendano et al., 2020), while one of the few closely related developing-country estimates comes from Kondirolli and Sunder (2022), who study school construction in Zimbabwe. Evidence for large school-construction programs in low- and middle-income countries remains sparse.

I study the long-run mental-health associations of Indonesia’s *Sekolah Dasar INPRES* program, which built approximately 61,800 primary schools between 1973 and 1978. Using IFLS-4 and IFLS-5, I relate cross-district differences in program intensity to CES-D outcomes through the interaction of district intensity with birth-cohort exposure. Because the INPRES instrument is weak for individual schooling in this sample, the paper focuses on reduced-form estimates and treats IV results as diagnostics.

21 The central empirical result is a negative female association. The pooled
22 estimates are also negative, but they are less credible because male pre-trends
23 fail. Geographic heterogeneity is not central to the paper: the appendix shows
24 that the Java/non-Java pattern is unstable and concentrated in IFLS-5 rather
25 than a stable pooled split.

26 First, I provide new reduced-form evidence on adult depressive symptoms
27 in the INPRES setting, complementing the broader quasi-experimental litera-
28 ture on education and depression and extending earlier reduced-form work on
29 INPRES and later-life health (Rizal et al., 2023). Second, I use the ten CES-D
30 items to suggest that the reduced-form symptom pattern differs from the stan-
31 dard cross-sectional education gradient. In OLS, education is most strongly
32 associated with positive-affect items. In the reduced form, the largest associa-
33 tions are with fearfulness and “could not get going.” That contrast motivates
34 the paper’s item-level analysis and frames the later discussion.

35 The remainder of the paper proceeds as follows. Section 2 describes the IN-
36 PRES program and the Indonesian mental-health context. Section 3 presents
37 the data and measurement. Section 4 sets out the identification strategy, includ-
38 ing a discussion of threats to the exclusion restriction and the weak-instrument
39 problem. Section 5 reports results. Section 6 interprets the findings. Section 7
40 concludes.

41 2. BACKGROUND

42 2.1. *The INPRES School Construction Program*

43 In 1973, the Indonesian government used oil-boom revenues to launch the
44 *Sekolah Dasar INPRES* program, which roughly doubled the number of primary
45 schools in the country by 1978/79 (Duflo, 2001). Allocation was redistributive,
46 and districts with lower initial enrollment rates received more schools. Chil-
47 dren born 1968–1972 were the primary exposed cohort, as they reached school
48 age after construction. Children born 1958–1962 were already enrolled when
49 construction began and serve as the comparison cohort.

50 District intensity varied because districts with lower initial enrollment re-
51 ceived more schools (Duflo, 2001). That redistribution creates the cross-district
52 variation used throughout the paper. The broader econometric concerns about
53 instrument strength and exclusion restrictions are taken up in Section 4, where
54 they matter directly for interpretation.

55 *2.2. Depression in Indonesia*

56 Depression is among the leading causes of disability worldwide, and Indone-
57 sia, the world’s fourth-most-populous country, faces a substantial and largely
58 unaddressed burden. Depressive symptoms may also affect labor supply, produc-
59 tivity, and household welfare. The 2018 national health survey (*Riskesdas*; Min-
60 istry of Health, Republic of Indonesia, 2018) estimated that 6.1% of adults ex-
61 experienced clinically significant depressive symptoms, though community-based
62 studies using validated screening instruments consistently report higher preva-
63 lence (Peltzer and Pengpid, 2018). In the analysis sample used here, 12.7%
64 screen positive using the standard CES-D 10 cutoff of ≥ 10 .

65 The treatment gap is severe, and specialist mental-health care remains lim-
66 ited (Lund et al., 2010; Patel et al., 2018). Cultural factors compound the
67 problem. Stigma around mental illness is widespread, and depressive symp-
68 toms are often expressed somatically rather than through the affective language
69 characteristic of Western clinical settings (Parker et al., 2001). This context
70 has two implications for the present study. First, if INPRES exposure is as-
71 sociated with lower depression, that association may reflect informal channels,
72 including reduced isolation, instead of formal treatment access, which was ef-
73 fectively nonexistent in rural areas during the INPRES period. None of these
74 channels is directly tested in this paper. Second, the CES-D 10, developed
75 by Andresen et al. (1994) and subsequently administered in Indonesian sam-
76 ples, includes both somatic and affective items and is well suited to capturing
77 depressive symptoms in this population.

78 2.3. *Related Literature*

79 This study relates to three literatures. The first asks what large school-
80 construction programs do in the long run. In Indonesia, that work has focused
81 mainly on education, labor markets, governance, and broader health outcomes
82 (Duflo, 2001, 2004; Akresh et al., 2023; Martinez-Bravo, 2017; Mazumder et al.,
83 2023; Rizal et al., 2023; Jung et al., 2021). Taken together, those papers suggest
84 that INPRES functioned as a large place-based intervention with consequences
85 that may extend beyond individual schooling.

86 The second literature studies education and depression. The causal evidence
87 is mixed and comes mostly from compulsory schooling reforms in high-income
88 countries. Some papers find protective effects (Crespo et al., 2014; Chevalier
89 and Feinstein, 2006), while others find no effect or adverse effects (Avendano
90 et al., 2020; Dahmann and Schnitzlein, 2019; Courtin et al., 2019). Outside
91 reform designs, sibling fixed-effects evidence also suggests that the education-
92 depression gradient is sensitive to selection (Frijters et al., 2014). Those designs
93 generally have stronger first stages than the INPRES setting, which is why the
94 Indonesian case is better suited to reduced-form program-exposure estimates
95 than to structural schooling returns.

96 The third literature is the small developing-country evidence base on school
97 expansion and mental health. Kondirolli and Sunder (2022) show that school
98 construction in Zimbabwe reduced depression among exposed cohorts. Beyond
99 providing a reduced-form estimate comparable to that work, this paper also
100 examines the CES-D response at the symptom level, which lets me compare the
101 program-exposure pattern with the standard cross-sectional education gradient
102 item by item.

103 3. DATA AND MEASUREMENT

104 3.1. *The Indonesian Family Life Survey*

105 The analysis pools the fourth and fifth waves of the Indonesian Family Life
106 Survey (IFLS-4 and IFLS-5), fielded in 2007/08 and 2014/15, respectively. The

IFLS is a longitudinal household survey with an individual tracking rate exceeding 92% across waves (Strauss et al., 2016). Pooling the two waves allows each individual to contribute up to two CES-D observations measured at different ages while leaving the identifying variation—birth cohort interacted with district intensity—unchanged. The pooled sample contains 60,474 person-wave CES-D observations. Restricting to the INPRES birth cohorts and imposing the control requirements yields the analysis sample of $N = 8,821$ person-wave observations from 5,144 unique individuals. Standard errors are clustered at the birth-district level throughout.

3.2. Depression Measurement: CES-D 10

The CES-D 10 (Andresen et al., 1994) is a 10-item screener measuring depressive symptom frequency over the past week, with each item rated 0–3. Total scores range from 0–30, with ≥ 10 as the standard screening cutoff. The ten items fall into four symptom groups:

- **Depressed affect:** Felt depressed (C), felt fearful (F), felt lonely (I)
- **Positive affect** (reverse-coded): Felt hopeful (E), felt happy (H)
- **Somatic symptoms:** Was bothered (A), had trouble concentrating (B), everything was an effort (D), sleep was restless (G)
- **Interpersonal:** Could not get “going” (J)

The IFLS coding is recoded to the standard 0–3 scale, with higher values indicating greater symptom severity. Coverage is nearly universal, with 99.98% of adult respondents having valid CES-D data.

3.3. Education and Control Variables

Controls include baseline district characteristics and individual demographics. District-level INPRES intensity (*nin*) measures schools built per 1,000 school-age children in 1971, and the baseline district controls are the 1971 water and sanitation index, enrollment rate, and population. The individual controls

are birth month, religion, ethnicity, language, and parental education. Current residence province fixed effects are excluded from the baseline because they may be post-treatment if INPRES affected migration; a later robustness check adds them back and does not reverse the main results. Table A1 provides full variable definitions.

3.4. Sample Construction

Table 1 summarizes sample construction, and Table A2 provides the detailed sample flow. Pooling IFLS-4 and IFLS-5, I start from 60,474 person-wave observations across all birth cohorts with valid CES-D scores, then restrict to the INPRES cohorts (born 1958–62 and 1968–72), yielding 10,104 observations. Imposing non-missing district-level and individual controls yields the analysis sample of $N = 8,821$ (4,282 male, 4,539 female). Restricting further to non-Java provinces gives $N = 3,692$ and to Java provinces, $N = 5,129$.

4. IDENTIFICATION STRATEGY

4.1. Reduced-Form Specification

The primary estimating equation is the reduced form, using the interaction of a “young” cohort indicator (born 1968–72) with district INPRES intensity:

$$\text{Depression}_{it} = \alpha_{d(i)} + \lambda_{b(i)} + \omega_t + \pi_1 \cdot (\text{young}_i \times \text{nin}_{d(i)}) + \mathbf{Z}'_{it}\boldsymbol{\psi} + \eta_{it} \quad (1)$$

where i indexes individuals and t indexes survey waves (IFLS-4 or IFLS-5). Here and below, Depression_{it} is a generic placeholder for either the continuous CES-D 10 score or the binary Depressed indicator, depending on the reported specification. $d(i)$ denotes the birth district of individual i , $b(i)$ denotes birth year, $\alpha_{d(i)}$ are birth-district fixed effects, $\lambda_{b(i)}$ are birth-year fixed effects, and ω_t are wave fixed effects. $\text{nin}_{d(i)}$ is INPRES intensity in that district, measured as schools built per 1,000 school-age children, and young_i indicates birth in 1968–1972 (the INPRES-exposed cohort). The coefficient π_1 captures the differential association of program intensity with depression between exposed and

unexposed cohorts. Cohorts born 1958–1962 serve as the comparison group, as they were already enrolled when INPRES began.

Birth-district fixed effects absorb time-invariant district characteristics, birth-year fixed effects absorb cohort differences, and wave fixed effects absorb survey-round differences. Identification comes from the interaction of district intensity with birth cohort: conditional on those fixed effects and controls, higher-intensity districts should show larger cohort differences than lower-intensity districts. The covariate vector $\mathbf{Z}_{it}$ includes baseline district trend interactions and individual demographics. The preferred specification adds those demographics to a baseline that already includes district trends.

For pre-trends testing, an event-study specification interacts district intensity with each birth-year indicator:

$$\text{Depression}_{it} = \alpha_{d(i)} + \lambda_{b(i)} + \omega_t + \sum_{\substack{y=1958 \\ y \neq 1962}}^{1972} \gamma_y \cdot (\text{nin}_{d(i)} \times \mathbf{1}[\text{birthyear}_i = y]) + \mathbf{Z}'_{it} \boldsymbol{\psi} + \eta_{it} \quad (2)$$

where $\mathbf{1}[\cdot]$ denotes the indicator function. The event-study specification allows each birth year to have a separate relationship with program intensity, making it possible to test whether pre-treatment cohorts exhibit differential trends across districts with different construction intensity. Detailed interpretation of those coefficients appears in the Results section.

4.2. Instrumental Variables Framework

For completeness, the analysis also reports IV estimates. The first stage is:

$$\text{Educ}_{it} = \alpha_{d(i)} + \lambda_{b(i)} + \omega_t + \sum_{\substack{y=1958 \\ y \neq 1962}}^{1972} \delta_y \cdot (\text{nin}_{d(i)} \times \mathbf{1}[\text{birthyear}_i = y]) + \mathbf{Z}'_{it} \boldsymbol{\zeta} + \varepsilon_{it} \quad (3)$$

and the structural second stage is:

$$\text{Depression}_{it} = \alpha_{d(i)} + \lambda_{b(i)} + \omega_t + \beta_1 \text{Educ}_{it} + \mathbf{Z}'_{it} \boldsymbol{\phi} + u_{it} \quad (4)$$

The reported 2SLS estimates use the predicted value of Educ_{it} from Equation (3). Equation (4) appears only to define the latent structural schooling

parameter β_1 , which is not the paper’s preferred estimand. I report 2SLS estimates, Kleibergen-Paap F -statistics, and Anderson-Rubin p -values only as diagnostics.

The appendix also reports the single-instrument first stage that exactly matches Eq. (1): it replaces the full set of birth-year–intensity interactions in Eq. (3) with the single term $\text{young}_i \times \text{nin}_{d(i)}$ and is estimated on the $N = 8,821$ INPRES-cohort sample (Table A3).

4.3. Threats to the Exclusion Restriction

The standard exclusion restriction requires that INPRES intensity affects depression only through education. Three concerns arise.

First, Martinez-Bravo (2017) shows that INPRES changed the educational composition of village leaders, who then improved local public goods provision (healthcare facilities, clean water). If these public goods independently affected mental health, the exclusion restriction fails.

Second, INPRES was part of a broader oil-boom development push. Water and sanitation programs were implemented concurrently, and districts receiving more schools may also have received more infrastructure investment, including health clinics (*Puskesmas*), roads, and electrification. The specification includes birth-year interactions with 1971 district characteristics (water/sanitation allocation, enrollment rate, population) to absorb differential trends correlated with construction intensity, but I cannot rule out residual confounding from time-varying district investments. The reduced-form estimates should therefore be interpreted as capturing the association of being in a high-intensity INPRES district, which may bundle school construction with correlated development investments, not the effect of school construction alone.

Third, INPRES may have induced selective migration: families in high-intensity districts may have migrated differently than those in low-intensity districts, and this migration could independently affect mental health decades later.

211 These concerns reinforce the case for treating the reduced form as the es-
 212 timand of interest. Under the weaker assumption that INPRES intensity is
 213 exogenous conditional on controls (parallel trends across districts with different
 214 construction intensity), the reduced-form coefficients capture the intent-to-treat
 215 association of the INPRES program with depression, regardless of whether the
 216 pathway runs through individual schooling, community-level changes, or cor-
 217 related investments. This is the interpretation I adopt throughout: I evaluate
 218 exposure to the INPRES program, not the return to a year of schooling.

219 4.4. *Weak Instruments*

220 The INPRES instrument is weak in this sample. In the event-study IV
 221 specification (Table 7), Kleibergen-Paap F -statistics are 2.40 for men and 0.86
 222 for women; the analogous single-instrument F -statistics (Table A3) are 3.43 and
 223 0.16 respectively. These values make conventional 2SLS inference unreliable. By
 224 the Lee et al. (2022) tF adjustment procedure, standard inference at $F = 2.40$
 225 requires a critical value of roughly 4.8 rather than 1.96. Because the reduced
 226 form remains well defined even when the first stage is weak (Chernozhukov
 227 and Hansen, 2008), and because INPRES likely bundled schooling with other
 228 local investments, the paper relies on reduced-form estimates for substantive
 229 inference and reports IV results only to show how little structural leverage the
 230 design provides.

231 4.5. *Identification Checks*

232 The placebo test uses pre-program cohorts (born 1950–1962), defining co-
 233 horts born 1957–1962 as “fake treatment” and 1950–1956 as “fake comparison.”
 234 Under the identifying assumption, INPRES intensity should show no association
 235 with depression differences among cohorts that were all too old to benefit. Bal-
 236 ance of predetermined district characteristics is tested against the instrument.
 237 Tables A4 and A5 report the corresponding balance and placebo results.

238 5. RESULTS

239 5.1. Descriptive Statistics

240 The pooled analysis sample has a mean CES-D 10 score of 4.84 (SD =
241 4.28), and 12.7% screen positive for depression (Table 2). Women report more
242 symptoms than men and have fewer years of schooling. About 59% of the sample
243 belongs to the young, INPRES-exposed cohort.

244 5.2. Reduced-Form Results

245 The reduced-form estimates are most persuasive in the female sample (Ta-
246 bles 3 and 4; Figure 1). Female pre-trend tests do not reject, though they are
247 low-powered, while the male pre-trends test is significant and the male placebo
248 estimate is marginally significant. I therefore treat the female estimates as pri-
249 mary, the pooled estimates as secondary summaries, and the male estimates as
250 descriptive only. This emphasis on women is not pre-specified; it is a credibility-
251 based prioritization that follows from the identifying checks.

252 In the preferred specification, the female CES-D coefficient is -0.510 (raw
253 $p = 0.001$; BH-adjusted $p = 0.007$), about 0.12 standard deviations of the CES-
254 D distribution. The pooled coefficient is also negative (-0.273 , raw $p = 0.016$;
255 BH-adjusted $p = 0.049$), but it combines the female pattern with male estimates
256 that do not satisfy the identifying tests. The male coefficient is smaller and
257 imprecise (-0.160 , $p = 0.390$). The binary outcome is directionally similar and
258 more precisely estimated for women than for the pooled sample: the female
259 Depressed coefficient is -0.029 , about a 21% reduction relative to the female
260 screening-positive rate reported in Table 3. Wild-cluster-bootstrap inference
261 reported below leaves the female CES-D estimate strongly significant and the
262 pooled CES-D estimate significant at conventional levels (Table A23).

263 The geographic results are best interpreted with caution. The appendix
264 shows that the more informative pattern is not a stable pooled contrast between
265 Java and non-Java, but a divergence that emerges only at the wave level. In
266 IFLS-4, the estimates are broadly similar across regions. In IFLS-5, by contrast,

the Java coefficients become positive while the non-Java coefficients become strongly negative, especially among women. But the positive Java estimate in IFLS-5 is not stable in leave-one-province-out exercises. Taken together, these patterns suggest heterogeneity across settings, but they do not support a clear or stable geographic contrast as a main result.

5.3. Item-Level Reduced-Form Decomposition

The item-level decomposition suggests that the reduced-form symptom pattern differs from the cross-sectional education gradient (Table 6; Figure A1). In OLS, the largest negative coefficients are on the positive-affect items, hopefulness and happiness. In the reduced form, those items are near zero. The strongest reduced-form association instead appears on fearfulness, and the only other pooled item surviving Benjamini-Hochberg correction is “could not get going.” The face-valid “felt depressed” item is near zero even though it still has a negative OLS education gradient in the same sample.

The female item pattern is directionally similar but sharper: sleep restless and fearfulness are among the most negative coefficients, while hopefulness and happiness remain near zero (Tables A6 and A7). A domain-level aggregation points the same way. Somatic, depressed-affect, and interpersonal domains are negative, while positive affect is near zero (Tables A8, A9, and A10). The pooled domain-equality tests are only suggestive, so the formal contrast is weaker than the item-level pattern itself. For women, the directional pattern is similar, but the cross-domain tests are too imprecise to support strong within-female claims.

5.4. IV Diagnostics (Reported for Completeness)

The IV estimates are retained only to show how little structural leverage the data provide. The event-study IV remains weak, with KP F -statistics of 0.12 for the pooled sample, 2.40 for men, and 0.86 for women (Table 7). At those values, 2SLS ratios are unstable and weak-instrument-robust Anderson–Rubin procedures do not recover informative confidence sets. The single-instrument IV results also support the findings (Tables A11 and A12). The positive pooled

296 2SLS coefficient is a mechanical ratio result because both the reduced form
 297 and the first stage are negative. The IV evidence therefore does not identify
 298 the return to schooling in this sample. It is reported only to document the
 299 weak-identification problem.

300 5.5. Robustness, Heterogeneity, and Multiple Testing

301 The appendix robustness checks refine the interpretation rather than over-
 302 turning it.

303 *Identification checks.* Figure 1 and Table 5 show that pre-treatment coef-
 304 ficients are near zero for the pooled sample and for women, while the male
 305 pre-trends test rejects. Tables A30 and A31 show that the female reduced-form
 306 association remains negative in both waves but the single-instrument first stage
 307 remains weak in each wave. Female treatment-cohort coefficients are uniformly
 308 negative, but the event study is much less precise than the single-instrument
 309 specification because it estimates five separate treatment effects on a larger sam-
 310 ple that includes the gap cohorts. Table A14 shows that later treatment cohorts
 311 matter for the pooled estimate but do not generate the female result mechan-
 312 ically. Table A13 also shows that the female pre-trends test is reassuring but
 313 low-powered.

314 *Specification checks.* The negative female CES-D association is already
 315 present in stripped-down specifications and changes little across alternative clus-
 316 tering choices, residence-province controls, one-observation-per-person samples,
 317 IFLS-5-only samples, alternative depression cutoffs, and Oster-style sensitivity
 318 calculations (Tables A16, A17, A18, A19, A21, and A20). Province-level clus-
 319 tering, leave-one-province-out exercises, and wild-cluster-bootstrap inference are
 320 more mixed for the pooled estimate than for the female estimate (Tables A22
 321 and A23). Dose-response evidence is suggestive rather than decisive: the most
 322 negative coefficients appear in the highest-intensity quartile, but the quartile
 323 pattern is not monotonic (Table A15). Table A32 shows that within-person
 324 fixed-effects regressions of depression on education are near zero, though those
 325 estimates are also limited by very little within-person schooling variation.

326 *Selection and sample composition.* Formal inclusion regressions in the exposure-
327 observed sample show no detectable relation between program intensity and en-
328 try into the final complete-case sample, and inverse-probability weighting leaves
329 the pooled and female reduced-form coefficients nearly unchanged (Tables A25
330 and A26). The same conclusion appears in the Oster calculations. Observable
331 differences by program intensity are mostly geographic rather than obviously
332 depression-related, though the female Muslim indicator is statistically distin-
333 guishable from zero (Tables A24, A27, and A4). The broader IFLS-1-linked
334 diagnostic also yields little evidence of stable treatment-related mortality or
335 pooled alive-but-not-observed attrition, though the marginally positive female
336 IFLS-5 attrition coefficient warrants caution (Tables A28 and A29).

337 *Heterogeneity and inference.* Geographic heterogeneity is better described
338 as an unstable IFLS-5 Java versus non-Java divergence than as a stable pooled
339 Java/non-Java split. I first estimate pooled Java/non-Java interaction mod-
340 els (Table A33), then separate the reduced form by wave (Tables A34, A36,
341 and A37), and finally test whether the IFLS-5 Java pattern survives leave-
342 one-province-out exercises (Table A35). Those checks all point in the same
343 direction: the pooled interaction tests reject equality, but the contrast is con-
344 centrated in IFLS-5, while the positive Java-side estimates weaken materially
345 once individual provinces are excluded. I therefore treat geography as a limit
346 on homogeneity rather than as evidence for a clean subgroup effect. Benjamini-
347 Hochberg correction is applied only to pre-defined inferential families: the six
348 headline reduced-form outcomes, the six IV Anderson-Rubin tests, the ten item
349 tests within each gender, and the four domain tests.

350 6. DISCUSSION

351 6.1. *The OLS-Reduced-Form Divergence*

352 A central descriptive result is the mismatch between the cross-sectional ed-
353 ucation gradient and the reduced-form symptom pattern. In OLS, education
354 is most strongly associated with lower scores on hopefulness and happiness. In

the reduced form, those items are near zero, while fearfulness and “could not get going” carry the largest negative coefficients. This contrast is clearest in the female sample, where the reduced-form evidence is most credible.

That contrast does not identify a mechanism. The OLS coefficient reflects the cross-sectional association per year of education and therefore mixes causal effects with selection. The reduced-form coefficient reflects exposure to the INPRES program and bundles schooling with community-level changes. The divergence is therefore consistent with several interpretations, including selection on optimism, program effects operating through non-schooling local channels, or omitted family-background factors that inflate the OLS gradient.

The domain-level aggregation points in the same direction, but less sharply than the item-level pattern. Positive affect remains near zero in the reduced form, while somatic, depressed-affect, and interpersonal domains are negative. The pooled domain-equality tests are only suggestive, and the female-only contrasts are too imprecise to support strong within-female claims. The formal tests therefore reinforce the item-level message without turning it into a decisive domain-level result.

6.2. Comparison with the Literature

The negative association here is consistent with Kondirolli and Sunder (2022) and Crespo et al. (2014), and it contrasts with Avendano et al. (2020), who study compulsory schooling in Britain. One possible institutional difference is that INPRES expanded access rather than imposing attendance, but many other cross-setting differences prevent a clean explanation of the sign contrast.

The preferred female estimate is modest in standard-deviation terms, about 0.12 of the pooled CES-D distribution. That is smaller than some estimates in the school-construction literature, but direct magnitude comparisons are difficult because the estimands, outcomes, and policy environments differ. The present estimate is a reduced-form association for a program not designed to target mental health.

The Java results should also remain secondary. The pooled interaction tests

385 reject equality, but the more informative pattern is wave-specific and unstable
386 within IFLS-5 Java. That makes geography more useful as a limit on homo-
387 geneity than as evidence for a well-estimated harmful effect in Java.

388 6.3. *Gender Patterns*

389 The gender pattern remains the paper’s main interpretive challenge. Women
390 show the strongest reduced-form association, but the female schooling first stage
391 is near zero. In this sample, program intensity is associated with later female
392 depression even though it is not detectably associated with female years of
393 schooling. That rules out a clean schooling-return interpretation, but it does
394 not identify an alternative mechanism.

395 The simplest implication is about the estimand. The paper identifies the as-
396 sociation of exposure to the INPRES program, not the return to a year of school-
397 ing. That association could reflect schooling, school quality, local public goods,
398 marriage-market effects, fertility responses, or other community-level changes.
399 Breierova and Duflo (2004) show that INPRES affected fertility, which is consis-
400 tent with the program having altered women’s life trajectories through channels
401 beyond own education and underscores why exclusion-restriction claims are not
402 credible here. The reduced-form results for women should therefore be read as
403 program-exposure associations rather than schooling effects.

404 6.4. *Limitations*

405 *Identification and estimand.* The weak first stage rules out credible struc-
406 tural estimates of the return to schooling. The design identifies exposure to a
407 high-intensity INPRES district, not education itself. That distinction is sharpest
408 for women, where the reduced-form result is strongest but the schooling first
409 stage is near zero. Bundled contemporaneous investments and possible spillovers
410 remain central threats, and direct measures of the relevant mechanisms are un-
411 available.

412 *Measurement.* CES-D 10 is a screening instrument rather than a clinical di-
413 agnosis, and it captures symptoms at the survey date rather than lifetime depres-
414 sion. Reverse-coded CES-D items and shortened CES-D batteries can display

415 unstable psychometric properties across settings (El-Ammari et al., 2023), so
416 measurement properties may still contribute to the item-level pattern. School-
417 ing is measured retrospectively, which may also amplify the divergence between
418 the OLS and reduced-form estimates.

419 *Selection and exposure measurement.* The new inclusion, weighting, and
420 IFLS-1-linked follow-up checks reduce concern about observable complete-case
421 selection and post-1993 mortality or attrition among tracked IFLS respondents,
422 but they do not eliminate deeper survival-selection concerns created by the long
423 lag between exposure and outcome measurement. Exposure assignment also de-
424 pends on retrospectively reported birth district linked to historical district codes,
425 and the INPRES intensity measure may contain non-classical measurement er-
426 ror. Spillovers across neighboring districts could also attenuate the measured
427 intensity contrast.

428 *External validity.* The estimates come from 13 Indonesian provinces and
429 may not generalize to eastern Indonesia or to other countries. The voluntary,
430 access-expanding character of INPRES also limits comparison with compulsory
431 schooling reforms. Depression is observed only in 2007 and 2014, so wave differ-
432 ences may reflect age, period, or cohort variation rather than a strengthening
433 association over time.

434 7. CONCLUSION

435 Under the parallel-trends assumption, exposure to high-intensity INPRES
436 districts is associated with lower adult CES-D scores decades later. The best-
437 supported reduced-form evidence remains in the female sample. That result
438 survives weighting, narrower treatment windows, and the new IFLS-1-linked
439 selection diagnostic. The pooled estimate is also negative, but because male pre-
440 trends fail, I treat it as a secondary summary rather than a credibly identified
441 estimate.

442 The paper’s main substantive contribution is the symptom-level contrast
443 between OLS and the reduced form. The reduced-form association is largest

444 for fearfulness and “could not get going,” not for the positive-affect items that
445 dominate the cross-sectional education gradient. That pattern suggests that
446 exposure to the INPRES program is linked to a narrower set of depressive-
447 symptom dimensions than the OLS education gradient would imply.

448 The claim therefore remains limited but substantive. The data do not iden-
449 tify the return to schooling or the mechanism behind the female result. They
450 do show that exposure to the INPRES program is associated with lower scores
451 on specific dimensions of adult depression, especially among women.

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454 All reported estimates, tables, and figures were checked against Stata output
455 generated from the replication scripts, and all interpretive decisions are my own.

456 **Conflict of Interest Statement**

457 The author declares no conflict of interest.

458 **Data Availability Statement**

459 The data used in this study come from the Indonesian Family Life Survey
460 (IFLS), which is publicly available from the RAND Corporation at [https://](https://www.rand.org/well-being/social-and-behavioral-policy/data/FLS/IFLS.html)
461 [www.rand.org/well-being/social-and-behavioral-policy/data/FLS/IFLS.](https://www.rand.org/well-being/social-and-behavioral-policy/data/FLS/IFLS.html)
462 [html](https://www.rand.org/well-being/social-and-behavioral-policy/data/FLS/IFLS.html). Replication code and do-files to reproduce all tables and figures will be
463 deposited in a public repository upon acceptance.

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Table 1: Sample Construction

Step	All	Male	Female
IFLS-4 respondents with CES-D	29,027	13,819	15,208
IFLS-5 respondents with CES-D	31,447	14,705	16,742
Pooled IFLS-4 + IFLS-5	60,474	28,524	31,950
Event-study sample (born 1958–72)	12,997	6,273	6,724
+ INPRES cohorts (born 1958–62 or 1968–72)	10,104	4,896	5,208
+ Non-missing controls	8,821	4,282	4,539
<i>Analysis sample by survey wave:</i>			
IFLS-4 (2007)	4,606	2,251	2,355
IFLS-5 (2014)	4,215	2,031	2,184
<i>Analysis sample by geography:</i>			
Java provinces	5,129	2,511	2,618
Non-Java provinces	3,692	1,771	1,921

Notes: Pooling IFLS-4 (2007) and IFLS-5 (2014) waves. INPRES cohorts follow the identification strategy of Duflo (2001): cohorts born 1968–72 (exposed to INPRES) and 1958–62 (comparison). Controls include birth district FE, birth year FE, wave FE, baseline district trend controls (birth year $\times$ 1971 water/sanitation, enrollment, population), religion, ethnicity, language, parental education, and birth month. Current residence province FE are excluded from the baseline specification and added only in Table A18. Java provinces: DKI Jakarta, West Java, Central Java, DI Yogyakarta, East Java, Banten (codes 31–36).

Table 2: Summary Statistics

	All		Male		Female	
	Mean	SD	Mean	SD	Mean	SD
<i>Panel A: Pooled IFLS-4 + IFLS-5 (All = 8,821; Male = 4,282; Female = 4,539)</i>						
CES-D 10 (0–30)	4.84	4.28	4.65	4.09	5.02	4.45
Depressed (≥ 10)	0.127	0.333	0.117	0.321	0.137	0.344
Education (years)	7.64	4.60	8.30	4.50	7.02	4.61
Young cohort (born 1968–72)	0.585	0.493	0.602	0.490	0.570	0.495
INPRES intensity (nin)	2.10	0.97	2.09	0.95	2.11	1.00
<i>Panel B: IFLS-4 Only (2007; All = 4,606; Male = 2,251; Female = 2,355)</i>						
CES-D 10 (0–30)	3.87	3.38	3.75	3.18	3.98	3.57
Depressed (≥ 10)	0.059	0.235	0.051	0.220	0.066	0.248
Education (years)	7.68	4.62	8.31	4.51	7.09	4.65
<i>Panel C: IFLS-5 Only (2014; All = 4,215; Male = 2,031; Female = 2,184)</i>						
CES-D 10 (0–30)	5.90	4.87	5.64	4.71	6.13	5.00
Depressed (≥ 10)	0.202	0.401	0.190	0.392	0.213	0.410
Education (years)	7.60	4.58	8.30	4.48	6.95	4.57

Notes: INPRES cohort analysis sample (born 1958–62 or 1968–72) with complete controls. Panel A pools both waves; Panels B and C show wave-specific means. CES-D 10 is the 10-item Center for Epidemiologic Studies Depression Scale (range 0–30). Depressed indicates CES-D ≥ 10 . INPRES intensity (nin) and young cohort shares are time-invariant and reported only in Panel A. Mean CES-D scores are higher in IFLS-5 (2014) than IFLS-4 (2007), consistent with age-related increases in depressive symptoms; wave fixed effects absorb these level differences.

Table 3: Reduced-Form: Association of Exposure to the INPRES Program with Depression (Full Sample)

	(1)	(2)
	Baseline	+ Demographics
<i>All</i> ($N = 8,821$)		
CES-D 10	-0.272** (0.112)	-0.273** (0.113)
Depressed	-0.016 (0.010)	-0.016 (0.010)
<i>Male</i> [†] ($N = 4,282$)		
CES-D 10	-0.145 (0.185)	-0.160 (0.185)
Depressed	-0.008 (0.015)	-0.008 (0.015)
<i>Female</i> ($N = 4,539$)		
CES-D 10	-0.511*** (0.155)	-0.510*** (0.155)
Depressed	-0.029 (0.014)	-0.029 (0.014)
District FE	Yes	Yes
Baseline district trends	Yes	Yes
Demographics	No	Yes

Notes: Each cell reports the coefficient on $\text{young}_i \times \text{nin}_{d(i)}$ from Eq. (1). Pooled IFLS-4 and IFLS-5 with wave fixed effects. Column (1): birth-year FE, district FE, and baseline district trends (birth year $\times$ 1971 water/sanitation, enrollment, population). Column (2): adds individual demographics. Standard errors clustered at the birth-district level in parentheses (209 clusters for All, 195 for Male, 187 for Female). Mean CES-D 10 is 4.84 (SD = 4.28) for All, 4.65 for Male, and 5.02 for Female; Depressed = 12.7% for All, 11.7% for Male, and 13.7% for Female. [†]Male pre-trends test is significant ($F = 3.03$, $p = 0.019$); male estimates should be interpreted with caution given the pre-trends failure (see Section 5.2). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All stars reflect Benjamini-Hochberg adjusted p -values, correcting for six primary hypotheses (3 samples $\times$ 2 outcomes).

Table 4: Reduced-Form: Java vs. Non-Java Provinces

	(1)	(2)
	Java	Non-Java
<i>All</i> ($N_{Java} = 5,129$; $N_{Non-Java} = 3,692$)		
CES-D 10	0.206 (0.184)	-0.557*** (0.149)
Depressed	0.021 (0.015)	-0.038*** (0.013)
<i>Male</i> [†] ($N_{Java} = 2,511$; $N_{Non-Java} = 1,771$)		
CES-D 10	0.348 (0.346)	-0.527* (0.267)
Depressed	0.031 (0.022)	-0.031 (0.023)
<i>Female</i> ($N_{Java} = 2,618$; $N_{Non-Java} = 1,921$)		
CES-D 10	-0.092 (0.339)	-0.712*** (0.216)
Depressed	0.009 (0.029)	-0.053*** (0.016)
District FE	Yes	Yes
Baseline district trends	Yes	Yes
Demographics	Yes	Yes

Notes: Each cell reports the coefficient on $\text{young}_i \times \text{nin}_{d(i)}$ from Eq. (1). Pooled IFLS-4 and IFLS-5 with wave fixed effects. Java provinces: codes 31–36. Both columns include district FE, birth-year FE, baseline district trends (birth year $\times$ 1971 water/sanitation, enrollment, population), and individual demographics (birth month, Muslim, Javanese, Indonesian speaker, parental education). Depressed = 1 if CES-D 10 $\geq$ 10. Standard errors clustered at the birth-district level in parentheses. Clusters: Java All = 100, Male = 100, Female = 96; Non-Java All = 109, Male = 95, Female = 91. [†]Male pre-trends test is significant ($F = 3.03$, $p = 0.019$); male estimates should be interpreted with caution given the pre-trends failure (see Section 5.2). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Stars reflect raw p -values (unlike Table 3, where stars reflect BH-adjusted p -values).

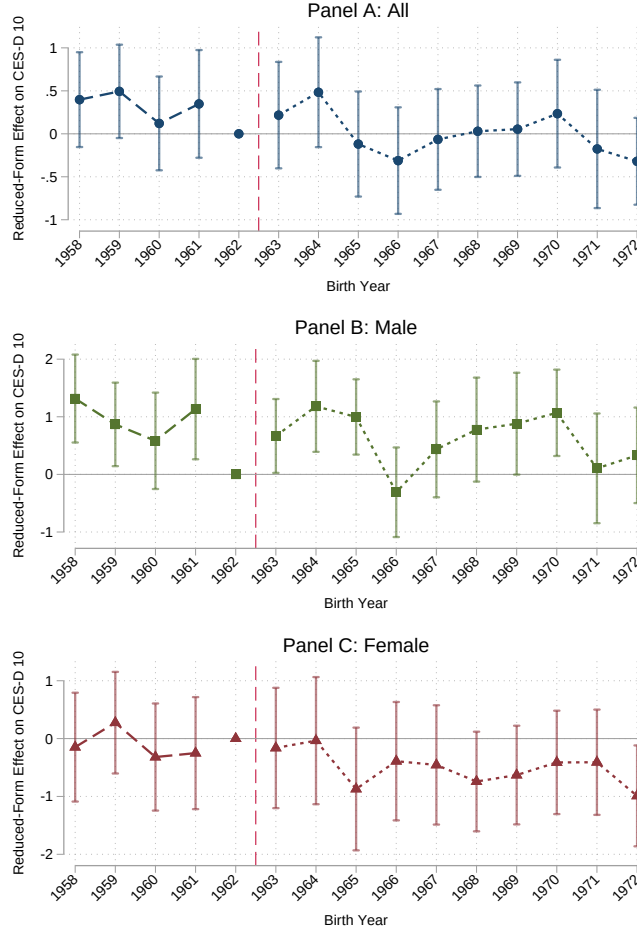


Figure 1: Reduced-Form Event Study: Association of Exposure to the INPRES Program with CES-D by Birth Year

Notes: The y -axis shows the reduced-form coefficient (CES-D 10 points) on $\text{nin}_{d(i)} \times \mathbf{1}[\text{birth year} = t]$ from a regression of CES-D 10 on birth-year–intensity interactions plus district FE, wave FE, baseline district controls, and individual demographics (religion, ethnicity, language, parental education, birth month). Reference year is 1962 (normalized to zero). The dashed line at 1962.5 marks the comparison-cohort boundary: pre-treatment coefficients (1958–1961) should be near zero under parallel trends; cohorts 1968–1972 are INPRES beneficiaries. Bars show 95% CIs clustered at the birth-district level. Pooled IFLS-4 and IFLS-5. The event-study specification includes all birth years 1958–1972 (including gap cohorts 1963–1967), yielding $N = 12,997$; the main INPRES-cohort analysis sample is $N = 8,821$. Top panel: pooled sample; middle panel: male; bottom panel: female. Clusters: 209 (All), 195 (Male), 187 (Female).

Table 5: Event-Study Coefficients for CES-D 10: Reduced-Form Coefficients by Birth Year

	All		Male		Female	
Birth year	Coef	SE	Coef	SE	Coef	SE
<i>Comparison cohort (reference: 1962)</i>						
1958	0.398	(0.281)	1.316***	(0.389)	−0.148	(0.480)
1959	0.494*	(0.277)	0.868**	(0.370)	0.275	(0.448)
1960	0.121	(0.278)	0.583	(0.427)	−0.319	(0.472)
1961	0.348	(0.320)	1.134**	(0.444)	−0.251	(0.494)
<i>Gap cohort (partial exposure)</i>						
1963	0.217	(0.316)	0.667**	(0.327)	−0.162	(0.530)
1964	0.484	(0.326)	1.181***	(0.403)	−0.035	(0.561)
1965	−0.119	(0.312)	0.997***	(0.333)	−0.871	(0.541)
1966	−0.312	(0.316)	−0.310	(0.397)	−0.390	(0.522)
1967	−0.065	(0.299)	0.435	(0.425)	−0.455	(0.526)
<i>Treatment cohort (INPRES beneficiaries)</i>						
1968	0.030	(0.271)	0.777*	(0.461)	−0.742*	(0.439)
1969	0.054	(0.278)	0.880*	(0.451)	−0.630	(0.435)
1970	0.234	(0.320)	1.069***	(0.382)	−0.411	(0.456)
1971	−0.176	(0.351)	0.105	(0.486)	−0.409	(0.464)
1972	−0.320	(0.258)	0.331	(0.423)	−0.990**	(0.445)
<i>Joint F-test: pre-treatment coefficients = 0 (1958–1961)</i>						
F-statistic	1.36		3.03**		0.81	
p-value	[0.248]		[0.019]		[0.522]	
<i>Joint F-test: treatment coefficients = 0 (1968–1972)</i>						
F-statistic	1.76		4.38***		1.36	
p-value	[0.122]		[0.001]		[0.240]	

Notes: $N = 12,997$ (All), 6,273 (Male), 6,724 (Female). Each cell reports the coefficient on $\text{nin}_{d(i)} \times \mathbf{1}[\text{birth year} = t]$ from the event-study reduced-form specification in Eq. (2), estimated on CES-D 10 with district FE, birth-year FE, wave FE, baseline district trends, and demographics. Reference year is 1962 (normalized to zero). Standard errors clustered by birth district (209/195/187). Stars reflect raw (unadjusted) p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Item-Level Decomposition: Reduced-Form vs. OLS

Item	Description	Reduced Form	OLS (Education)
<i>Somatic symptoms</i>			
A	Bothered	−0.027 (0.018)	−0.000 (0.002)
B	Concentrating	−0.053 (0.028)	0.001 (0.002)
D	Effort	−0.057 (0.033)	0.001 (0.003)
G	Sleep	−0.040 (0.026)	−0.015*** (0.003)
<i>Depressed affect</i>			
C	Depressed	0.014 (0.020)	−0.006*** (0.002)
F	Fearful	−0.069*** (0.022)	−0.007*** (0.002)
I	Lonely	−0.037 (0.017)	−0.009*** (0.002)
<i>Positive affect (reverse-coded)</i>			
E	Hopeful (R)	0.033 (0.034)	−0.036*** (0.003)
H	Happy (R)	0.013 (0.030)	−0.025*** (0.003)
<i>Interpersonal</i>			
J	Could not get going	−0.052*** (0.019)	−0.006*** (0.002)
<i>N</i>		8,821	

Notes: Each row reports the coefficient on $\text{young}_i \times \text{nin}_{d(i)}$ (reduced form) or years of education (OLS) for the listed CES-D item (0–3 scale). Controls include district FE, birth-year FE, wave FE, baseline district trends, and demographics. Items are grouped by CES-D dimension; (R) = reverse-coded. Standard errors clustered by birth district. Stars in the reduced-form column reflect Benjamini-Hochberg adjusted p -values across 10 item tests; stars in the OLS column reflect raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: Weak-Instrument Diagnostics and IV Estimates (Reported for Completeness)

	2SLS	KP F	BH-adj AR p
<i>All</i> ($N = 8,821$)			
CES-D 10	1.948 (2.779)	0.12	0.027
Depressed	0.095 (0.138)	0.12	0.127
<i>Male</i> [†] ($N = 4,282$)			
CES-D 10	-0.450 (0.304)	2.40	0.027
Depressed	-0.046 (0.023)	2.40	0.099
<i>Female</i> ($N = 4,539$)			
CES-D 10	-0.908 (0.487)	0.86	0.027
Depressed	-0.040 (0.030)	0.86	0.283

Notes: Pooled IFLS-4 and IFLS-5 with wave FE. 2SLS estimates reported for completeness using the event-study specification with birth-year $\times$ district intensity interactions as instruments (Eq. (3)). Controls: district FE, birth-year FE, wave FE, baseline district trends (birth year $\times$ 1971 water/sanitation, enrollment, population), and individual demographics (birth month, religion, ethnicity, language, parental education), as in Table 3 Column (2). Standard errors clustered at the birth-district level in parentheses (209 clusters for All, 195 for Male, 187 for Female). KP F = Kleibergen-Paap F -statistic. BH-adj AR p = Benjamini-Hochberg adjusted Anderson-Rubin p -values, correcting for 6 hypotheses (3 samples $\times$ 2 outcomes). Anderson-Rubin confidence sets are unbounded in all specifications because $F < 3.84$, confirming that the instrument is too weak for structural inference. Stars omitted: standard inference is unreliable at these F -statistics. At $F = 2.40$, the Lee et al. (2022) tF -adjusted critical value is ≈ 4.8 . [†]Male estimates do not pass pre-trends tests ($F = 3.03$, $p = 0.019$) and should be interpreted with caution. The main evidence is the reduced form (Tables 3–4).

Table A1: Variable Definitions

Variable	Type	Definition
<i>Outcome variables</i>		
CES-D 10	Continuous	10-item CES-D scale (range 0–30). Higher = more depressive symptoms.
Depressed	Binary	= 1 if CES-D 10 $\geq$ 10
<i>Education</i>		
Years of education	Continuous	Completed years of schooling
<i>Instruments</i>		
$\text{nin}_{d(i)}$	Continuous	Number of INPRES schools constructed per 1,000 school-age children in individual i 's birth district d (1971 census)
Young	Binary	= 1 if born 1968–1972 (treatment cohort)
<i>District-level controls (1971 census)</i>		
wsppc	Continuous	Water and sanitation program allocation per capita
en71	Continuous	School enrollment rate in 1971
pop71	Continuous	District population in 1971
<i>Individual-level controls</i>		
Birth year	FE	Year of birth (1958–1972 for INPRES cohorts)
Birth month	Continuous	Month of birth (1–12)
Birth district	FE	District of birth (propkab)
Muslim	Binary	= 1 if religion is Islam
Javanese	Binary	= 1 if ethnicity is Javanese
Language	Binary	= 1 if Indonesian spoken at home
Mother's education	Binary	= 1 if mother completed primary school
Father's education	Binary	= 1 if father completed primary school

Notes: All variables constructed from IFLS-4/5 unless otherwise noted. District-level instruments from Dufo (2001) replication data. CES-D items are kp02a–kp02j in the IFLS mental health module.

Table A2: Detailed Sample Construction

Step	All	Male	Female
<i>Panel A: Full pooled sample</i>			
IFLS-4 adult respondents with CES-D	29,027	13,819	15,208
IFLS-5 adult respondents with CES-D	31,447	14,705	16,742
Pooled IFLS-4 + IFLS-5	60,474	28,524	31,950
Non-missing individual controls	56,624	26,772	29,852
<i>Panel B: INPRES cohort restriction</i>			
Born 1958–1962 or 1968–1972	10,104	4,896	5,208
Non-missing district and individual controls	8,821	4,282	4,539
<i>Panel C: Non-Java restriction</i>			
Non-Java provinces only	3,692	1,771	1,921

Notes: Panels A–C show the pooled IFLS-4 (2007) and IFLS-5 (2014) sample construction. Panel B restricts to INPRES birth cohorts (born 1958–62 and 1968–72) with complete controls ($N = 8,821$). Panel C excludes Java provinces (codes 31–36).

Table A3: First-Stage Results (Single Instrument: $\text{young} \times \text{nin}$)			
Outcome	(1)	(2)	(3)
	Birth-year + District FE	+ District trends	+ Individual demographics
<i>All</i> ($N = 8,821$)			
Years of education	−0.030 (0.122)	−0.141 (0.160)	−0.158 (0.144)
<i>F</i> -statistic	0.06	0.78	1.20
<i>Male</i> [†] ($N = 4,282$)			
Years of education	−0.133 (0.135)	−0.354* (0.190)	−0.340* (0.183)
<i>F</i> -statistic	0.97	3.49	3.43
<i>Female</i> ($N = 4,539$)			
Years of education	0.059 (0.154)	0.126 (0.216)	0.078 (0.197)
<i>F</i> -statistic	0.15	0.34	0.16

Notes: First-stage regression of years of education on $\text{young}_i \times \text{nin}_{d(i)}$ (single instrument). Pooled IFLS-4 and IFLS-5 with wave fixed effects. Standard errors clustered at the birth-district level in parentheses. Specifications: (1) birth-year, wave, and district fixed effects only; (2) adds district trend controls (birth year $\times$ 1971 water/sanitation, enrollment, population); (3) adds individual demographics (birth month, religion, ethnicity, language, parental education). The baseline specification excludes current residence province fixed effects because they may be post-treatment; Table A18 adds them as a robustness check. Reported *F*-statistics are the first-stage test statistics on $\text{young}_i \times \text{nin}_{d(i)}$. Cluster counts: 209 (All), 195 (Male), 187 (Female). Significance levels are based on unadjusted *p*-values. Main-text Table 3 reports BH-adjusted stars for the six headline reduced-form outcomes; most appendix tables report raw *p*-values unless otherwise noted. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4: Balance Tests (Single Instrument)

	(1)	(2)	(3)
Dependent variable	All	Male	Female
Muslim	−0.014 (0.010)	−0.006 (0.022)	−0.023** (0.011)
Bahasa speaker	0.001 (0.006)	−0.008 (0.011)	0.005 (0.009)
Javanese ethnicity	−0.003 (0.009)	−0.009 (0.012)	−0.002 (0.014)
Mother educated	−0.003 (0.018)	0.021 (0.027)	−0.019 (0.025)
Father educated	0.009 (0.014)	−0.011 (0.021)	0.037* (0.021)
Birth month	0.244* (0.145)	0.216 (0.178)	0.265 (0.187)
<i>N</i>	8,821	4,282	4,539

Notes: Each cell reports the coefficient on $\text{young} \times \text{nin}$ from a regression of the indicated predetermined characteristic on the instrument with full controls (excluding the dependent variable from controls to avoid mechanical correlation). Pooled IFLS-4 and IFLS-5 with wave fixed effects. INPRES cohorts (born 1958–62, 1968–72). Standard errors clustered at the birth-district level (209 clusters for All, 195 for Male, 187 for Female) in parentheses. Of 18 balance tests, one rejects at 5% (Muslim for females); father’s education for females rejects at 10%—consistent with the expected false positive rate. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A5: Placebo Tests: Pre-Program Cohorts (Single Instrument)

	(1)	(2)	(3)
Outcome	All	Male [†]	Female
<i>Panel A: CES-D 10</i>			
young $\times$ nin	-0.200	-0.496*	-0.159
	(0.173)	(0.254)	(0.430)
<i>Panel B: Depressed (≥ 10)</i>			
young $\times$ nin	0.003	-0.011	-0.000
	(0.017)	(0.024)	(0.027)
<i>N</i>	3,657	1,704	1,953

Notes: Pooled IFLS-4 and IFLS-5 with wave fixed effects. Placebo tests use only pre-program cohorts (born 1950–1962), defining “fake treatment” (1957–1962) vs. comparison (1950–1956). Full controls: district FE, birth-year FE, wave FE, baseline district trend controls (birth year $\times$ 1971 water/sanitation, enrollment, population), and individual demographics (birth month, religion [Muslim], ethnicity [Javanese], language [Indonesian speaker], parental education [mother and father]). Standard errors clustered at the birth-district level (209 clusters for All, 195 for Male, 187 for Female) in parentheses. The pooled and female placebo coefficients are close to zero for both outcomes; the male CES-D placebo remains the notable exception. [†]Male estimates do not pass pre-trends tests ($F = 3.03$, $p = 0.019$); the male placebo coefficient is marginally significant ($p = 0.053$), further warranting caution. Significance levels are based on unadjusted p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A6: Item-Level Reduced-Form Coefficients by Gender

Item	Description	All	Male [†]	Female
<i>Somatic symptoms</i>				
A	Was bothered	−0.027 (0.018)	−0.003 (0.027)	−0.053** (0.026)
B	Trouble concentrating	−0.053* (0.028)	−0.048 (0.042)	−0.061* (0.036)
D	Everything was an effort	−0.057* (0.033)	−0.050 (0.047)	−0.083 (0.050)
G	Sleep was restless	−0.040 (0.026)	0.013 (0.036)	−0.098** (0.044)
<i>Depressed affect</i>				
C	Felt depressed	0.014 (0.020)	0.022 (0.032)	−0.008 (0.032)
F	Felt fearful	−0.069*** (0.022)	−0.060* (0.034)	−0.088*** (0.030)
I	Felt lonely	−0.037** (0.017)	−0.030 (0.022)	−0.054** (0.025)
<i>Positive affect (reverse-coded)</i>				
E	Felt hopeful about future	0.033 (0.034)	0.020 (0.049)	0.024 (0.041)
H	Was happy	0.013 (0.030)	0.015 (0.049)	−0.019 (0.044)
<i>Interpersonal</i>				
J	Could not get going	−0.052*** (0.019)	−0.039 (0.026)	−0.070** (0.028)
<i>N</i>		8,821	4,282	4,539

Notes: Each cell reports the reduced-form coefficient from regressing the individual CES-D item (scored 0–3) on young $\times$ nin with full controls. Pooled IFLS-4 and IFLS-5 with wave fixed effects. Standard errors clustered at the birth-district level in parentheses. Positive affect items (E, H) are reverse-coded. INPRES cohorts (born 1958–62, 1968–72). [†]Male pre-trends test is significant (see Table 3 notes for details). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Significance levels are based on Benjamini-Hochberg adjusted p -values within each column (10 item-level hypotheses).

Table A7: Item-Level OLS Associations of Education with Depression by Gender

Item	Description	All	Male	Female
<i>Somatic symptoms</i>				
A	Was bothered	−0.000 (0.002)	0.002 (0.003)	0.001 (0.004)
B	Trouble concentrating	0.001 (0.002)	0.000 (0.003)	0.005 (0.004)
D	Everything was an effort	0.001 (0.003)	−0.003 (0.005)	0.002 (0.004)
G	Sleep was restless	−0.015*** (0.003)	−0.008* (0.005)	−0.017*** (0.005)
<i>Depressed affect</i>				
C	Felt depressed	−0.006*** (0.002)	−0.004 (0.003)	−0.006** (0.003)
F	Felt fearful	−0.007*** (0.002)	−0.002 (0.003)	−0.008** (0.004)
I	Felt lonely	−0.009*** (0.002)	−0.009*** (0.003)	−0.009*** (0.003)
<i>Positive affect (reverse-coded)</i>				
E	Felt hopeful about future	−0.036*** (0.003)	−0.034*** (0.005)	−0.039*** (0.005)
H	Was happy	−0.025*** (0.003)	−0.026*** (0.005)	−0.025*** (0.005)
<i>Interpersonal</i>				
J	Could not get going	−0.006*** (0.002)	−0.006** (0.003)	−0.005* (0.003)
N		8,821	4,282	4,539

Notes: Each cell reports the OLS coefficient on years of education from a regression of the individual CES-D item (scored 0–3) with full controls. Regressor: years of education. Pooled IFLS-4 and IFLS-5 with wave fixed effects. Standard errors clustered at the birth-district level in parentheses. Positive affect items (E, H) are reverse-coded. OLS significance based on raw (unadjusted) p -values. See Table A6 for BH-adjusted reduced-form significance. INPRES cohorts (born 1958–62, 1968–72). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

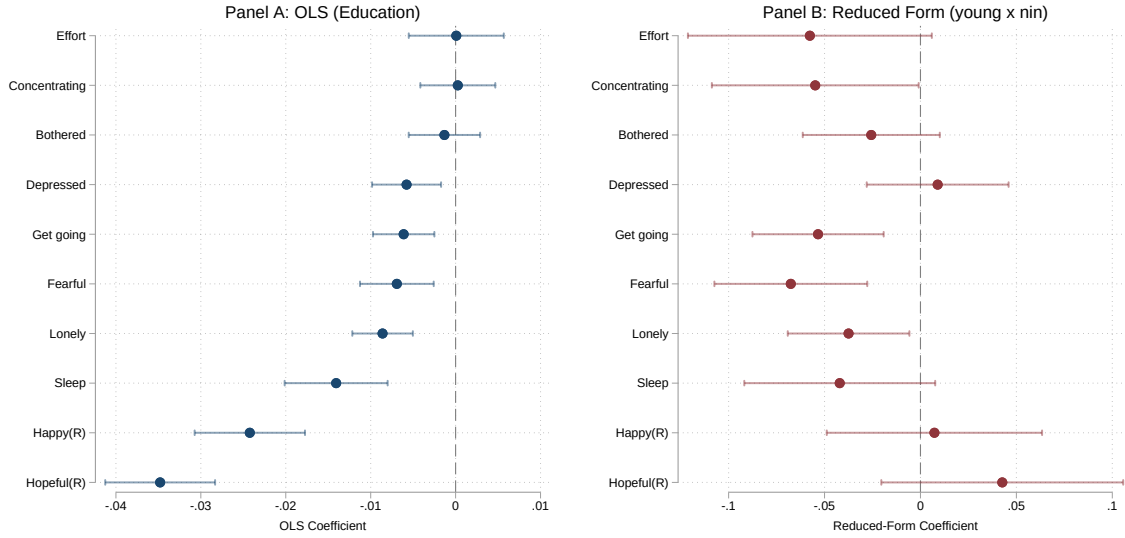


Figure A1: CES-D Item-Level Coefficients: OLS vs. Reduced-Form

Notes: The y -axis shows the estimated coefficient (item score, 0–3 scale). Panel A plots the OLS coefficient of years of education and Panel B the reduced-form coefficient of young $\times$ nin on each CES-D item, with 95% confidence intervals based on raw cluster-robust standard errors. Items are grouped by CES-D domain (Table A8): Somatic (A, B, D, G), Depressed affect (C, F, I), Positive affect (E, H), and Interpersonal (J). Positive affect items (hopeful, happy) are reverse-coded so negative coefficients indicate fewer symptoms. The OLS–reduced-form divergence is visible: OLS associates education most strongly with positive affect, but the largest pooled reduced-form coefficients are on fearfulness and “could not get going,” with positive-affect items indistinguishable from zero. Pooled IFLS-4 and IFLS-5 sample ($N = 8,821$) with full controls. Standard errors clustered at the birth-district level (209 clusters).

Table A8: Domain-Level Reduced-Form Coefficients

Domain	All		Male [†]		Female	
	Coef	<i>p</i>	Coef	<i>p</i>	Coef	<i>p</i>
Somatic (A+B+D+G)	−0.176*** (0.064)	0.006	−0.088 (0.100)	0.384	−0.295*** (0.094)	0.002
Depressed affect (C+F+I)	−0.091** (0.045)	0.045	−0.068 (0.069)	0.330	−0.150** (0.066)	0.024
Positive affect (E+H)	0.046 (0.053)	0.390	0.035 (0.082)	0.675	0.005 (0.066)	0.939
Interpersonal (J)	−0.052*** (0.019)	0.008	−0.039 (0.026)	0.126	−0.070** (0.028)	0.012
<i>N</i>	8,821		4,282		4,539	

Notes: Reduced-form coefficients on $\text{young}_i \times \text{nin}_{d(i)}$ with full controls. Domain indices follow the standard CES-D psychometric structure (Andresen et al., 1994): Somatic (bothered, concentrating, effort, sleep restless), Depressed affect (depressed, fearful, lonely), Positive affect (hopeful, happy), and Interpersonal (could not get going). This grouping is pre-specified based on the established CES-D factor structure, not derived from the data. The domain scores in this table are sums of the underlying items; Table A9 reports the corresponding average-domain coefficients used in the formal cross-domain tests. Positive affect items are reverse-coded in the CES-D, so higher values indicate *less* positive affect. Benjamini-Hochberg adjustment is applied across 4 domain tests within each sample. In the pooled sample, the adjusted *p*-values are 0.016 for Somatic, 0.060 for Depressed affect, 0.390 for Positive affect, and 0.016 for Interpersonal. In the female sample, the adjusted *p*-values are 0.008 for Somatic, 0.032 for Depressed affect, 0.939 for Positive affect, and 0.024 for Interpersonal. Standard errors clustered at the birth-district level (209 clusters for All, 195 for Male, 187 for Female) are in parentheses. [†]Male pre-trends test is significant (see Table 3 notes for details). Stars reflect raw (unadjusted) *p*-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A9: Formal Domain-Level Coefficients

Domain	All	Male	Female
<i>Panel A: Reduced form</i>			
Somatic	−0.0441*** (0.0159)	−0.0219 (0.0251)	−0.0737*** (0.0235)
Depressed affect	−0.0304** (0.0151)	−0.0225 (0.0230)	−0.0499** (0.0219)
Positive affect	0.0231 (0.0267)	0.0173 (0.0412)	0.0025 (0.0329)
Interpersonal	−0.0516*** (0.0193)	−0.0393 (0.0256)	−0.0703** (0.0277)
<i>Panel B: OLS (education)</i>			
Somatic	−0.0032* (0.0017)	−0.0023 (0.0024)	−0.0025 (0.0029)
Depressed affect	−0.0074*** (0.0016)	−0.0050** (0.0021)	−0.0077*** (0.0023)
Positive affect	−0.0301*** (0.0026)	−0.0303*** (0.0038)	−0.0318*** (0.0037)
Interpersonal	−0.0063*** (0.0018)	−0.0057** (0.0026)	−0.0053* (0.0027)

Notes: Domain scores average the CES-D items within somatic (4 items), depressed-affect (3 items), positive-affect (2 items, reverse-coded), and interpersonal (1 item) groupings. $N = 8,821$ (All), 4,282 (Male), 4,539 (Female). All regressions use the preferred reduced-form specification with district FE, birth-year FE, wave FE, baseline district trends, and demographics. Standard errors clustered at the birth-district level (209 clusters for All, 195 for Male, 187 for Female) are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A10: Formal Cross-Domain Equality Tests

Test	All	Male	Female
<i>Panel A: Reduced form</i>			
All four equal, $\chi^2(3)$	5.278 (0.1525)	1.417 (0.7015)	4.112 (0.2496)
Positive = Somatic, $\chi^2(1)$	3.842 (0.0500)	0.676 (0.4109)	3.321 (0.0684)
Positive = Depressed affect, $\chi^2(1)$	3.241 (0.0718)	0.784 (0.3759)	1.627 (0.2021)
Positive = Interpersonal, $\chi^2(1)$	4.998 (0.0254)	1.391 (0.2382)	2.430 (0.1191)
<i>Panel B: OLS (education)</i>			
All four equal, $\chi^2(3)$	81.352 (0.0000)	42.167 (0.0000)	44.912 (0.0000)
Positive = Somatic, $\chi^2(1)$	75.512 (0.0000)	40.120 (0.0000)	38.828 (0.0000)
Positive = Depressed affect, $\chi^2(1)$	59.890 (0.0000)	38.015 (0.0000)	32.107 (0.0000)
Positive = Interpersonal, $\chi^2(1)$	67.081 (0.0000)	28.019 (0.0000)	39.567 (0.0000)

Notes: Chi-squared tests of equality across domain-specific coefficients. The reduced-form panel tests whether the program-exposure pattern differs across CES-D domains; the OLS panel provides the corresponding education-gradient comparison. $N = 8,821$ (All), 4,282 (Male), 4,539 (Female). All regressions use the preferred specification with district FE, birth-year FE, wave FE, baseline district trends, and demographics; standard errors clustered at the birth-district level (209 clusters for All, 195 for Male, 187 for Female). p -values are reported in parentheses.

Table A11: Single Instrument 2SLS Estimates: $\text{young} \times \text{nin}$

	(1)
Outcome	$\hat{\beta}_{2\text{SLS}}$
<i>All</i> ($N = 8,821$; $F = 1.20$)	
CES-D 10	1.731
	(1.714)
Depressed	0.102
	(0.119)
<i>Male</i> [†] ($N = 4,282$; $F = 3.43$)	
CES-D 10	0.470
	(0.600)
Depressed	0.024
	(0.047)
<i>Female</i> ($N = 4,539$; $F = 0.16$)	
CES-D 10	−6.502
	(15.517)
Depressed	−0.374
	(0.896)

Notes: 2SLS estimates using $\text{young} \times \text{nin}$ as the single instrument for years of education. Pooled IFLS-4 and IFLS-5 with wave fixed effects. Full controls. Standard errors clustered at the birth-district level in parentheses (209 clusters for All, 195 for Male, 187 for Female). F = first-stage Kleibergen-Paap F -statistic. Stars omitted: all F -statistics are below the conventional threshold of 10, making standard inference unreliable. Anderson-Rubin p -values are reported in Table 7 of the main text. INPRES cohorts (born 1958–62, 1968–72).

Table A12: Anderson-Rubin Confidence Sets for the Structural Schooling Parameter

Sample	Outcome	AR 95% confidence set
All	CES-D 10	Unbounded
All	Depressed (≥ 10)	Unbounded
Male [†]	CES-D 10	Unbounded
Male [†]	Depressed (≥ 10)	Unbounded
Female	CES-D 10	Unbounded
Female	Depressed (≥ 10)	Unbounded

Notes: Anderson-Rubin confidence sets are computed for the structural coefficient on years of schooling using the same event-study IV specification reported in Table 7. In all six cases the confidence set is unbounded, which is another way of stating that the instrument is too weak to deliver informative structural inference. [†]Male estimates do not pass pre-trends tests and are reported only for completeness.

Table A13: Pre-Treatment Joint Tests and Common-Shift Minimum Detectable Effects

Statistic	All	Male	Female
<i>Panel A: Outcome = CES-D 10</i>			
N	12,997	6,273	6,724
Pre-trend F	1.3617	3.0278	0.8070
Pre-trend p	0.2483	0.0187	0.5221
MDE80 common shift	0.813	1.083	1.370
<i>Panel B: Outcome = Depressed</i>			
N	12,997	6,273	6,724
Pre-trend F	0.9271	2.3078	0.8789
Pre-trend p	0.4491	0.0594	0.4776
MDE80 common shift	0.066	0.088	0.110

Notes: The minimum detectable effect is the smallest common shift across the four pre-treatment cohort coefficients that would deliver 80 percent power against the 5 percent joint test threshold in the event-study specification.

Table A14: Cohort-Window Sensitivity for the Preferred Reduced Form

Treatment window	All	Male	Female
<i>Panel A: Outcome = CES-D 10</i>			
1968–1972 ($N = 8,821/4,282/4,539$)	−0.2733** (0.1129)	−0.1597 (0.1854)	−0.5099*** (0.1547)
1969–1972 ($N = 7,832/3,806/4,026$)	−0.2802** (0.1241)	−0.1541 (0.1770)	−0.4820*** (0.1546)
1970–1972 ($N = 6,847/3,311/3,536$)	−0.3085* (0.1576)	−0.2286 (0.2241)	−0.4884*** (0.1624)
1971–1972 ($N = 5,798/2,770/3,028$)	−0.4917*** (0.1771)	−0.5614** (0.2603)	−0.6244*** (0.1752)
<i>Panel B: Outcome = Depressed</i>			
1968–1972 ($N = 8,821/4,282/4,539$)	−0.0161 (0.0101)	−0.0082 (0.0150)	−0.0294** (0.0136)
1969–1972 ($N = 7,832/3,806/4,026$)	−0.0165 (0.0113)	−0.0124 (0.0173)	−0.0253* (0.0132)
1970–1972 ($N = 6,847/3,311/3,536$)	−0.0200 (0.0130)	−0.0215 (0.0179)	−0.0252* (0.0135)
1971–1972 ($N = 5,798/2,770/3,028$)	−0.0329** (0.0145)	−0.0433** (0.0211)	−0.0369** (0.0163)

Notes: Each row redefines the treatment cohort to births in the listed upper window and keeps the control cohort at 1958–1962. All coefficients come from the preferred reduced-form specification with the paper’s full control set. Standard errors clustered at the birth-district level are in parentheses. Significance stars are based on raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A15: Dose-Response in CES-D 10 by INPRES Intensity Quartile

INPRES intensity quartile	All		Male [†]		Female	
	Coef	SE	Coef	SE	Coef	SE
Q1 (low; mean nin = 0.88)	[ref.]		[ref.]		[ref.]	
Q2 (mean nin = 1.67)	−0.348	(0.242)	−0.676*	(0.355)	−0.162	(0.378)
Q3 (mean nin = 2.16)	−0.309	(0.256)	−0.263	(0.400)	−0.581	(0.389)
Q4 (high; mean nin = 3.35)	−0.673**	(0.299)	−0.696	(0.445)	−1.042**	(0.473)
<i>N</i>	8,821		4,282		4,539	

Notes: Dependent variable: CES-D 10 score. Each cell reports the coefficient on $\text{young}_i \times \mathbf{1}[\text{nin}_{d(i)} \in Q_k]$ from a regression of CES-D 10 on quartile-cohort interactions plus district FE, birth-year FE, wave FE, baseline district trend controls, and individual demographics (birth month, religion [Muslim], ethnicity [Javanese], language [Indonesian speaker], parental education [mother and father]). Q1 (lowest INPRES intensity) is the omitted reference category. Quartile cutpoints are based on the district-level distribution of nin (number of INPRES schools constructed per 1,000 children). Pooled IFLS-4 and IFLS-5. Standard errors clustered at the birth-district level (209 clusters for All, 195 for Male, 187 for Female) in parentheses. [†]Male pre-trends test is significant (see Table 3 notes for details). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A16: Reduced-Form Estimates Across Nested Control Sets

	(1)	(2)	(3)
	Sparse FE only	+ District trends	+ Demographics
<i>Panel A: Outcome = CES-D 10</i>			
All ($N = 8,821$)	−0.246*** (0.089)	−0.272*** (0.112)	−0.273*** (0.113)
Male [†] ($N = 4,282$)	−0.094 (0.156)	−0.145 (0.185)	−0.160 (0.185)
Female ($N = 4,539$)	−0.482*** (0.118)	−0.511*** (0.155)	−0.510*** (0.155)
<i>Panel B: Outcome = Depressed (≥ 10)</i>			
All ($N = 8,821$)	−0.013 (0.008)	−0.016* (0.010)	−0.016* (0.010)
Male [†] ($N = 4,282$)	−0.008 (0.013)	−0.008 (0.015)	−0.008 (0.015)
Female ($N = 4,539$)	−0.021** (0.010)	−0.029** (0.014)	−0.029** (0.014)

Notes: Each cell reports the reduced-form coefficient on $\text{young}_i \times \text{nin}_{d(i)}$ from Eq. (1). Pooled IFLS-4 and IFLS-5 with wave fixed effects. Column (1) includes only district, birth-year, and wave fixed effects. Column (2) adds baseline district trend controls (birth year $\times$ 1971 water/sanitation, enrollment, population). Column (3) adds individual demographics (birth month, religion, ethnicity, language, parental education) and is the preferred baseline specification. Current residence province fixed effects are excluded here because they may be post-treatment; Table A18 adds them as a robustness check. Standard errors clustered at the birth-district level in parentheses. [†]Male estimates do not pass pre-trends tests and are reported only for completeness. Significance levels are based on raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A17: Standard Error Sensitivity: Alternative Clustering

	(1)	(2)	(3)
	Robust	District	Province
<i>Panel A: Outcome = CES-D 10</i>			
All ($N = 8,821$)	-0.273** (0.131)	-0.273*** (0.113)	-0.273** (0.120)
Male [†] ($N = 4,282$)	-0.160 (0.196)	-0.160 (0.185)	-0.160 (0.210)
Female ($N = 4,539$)	-0.510*** (0.184)	-0.510*** (0.155)	-0.510*** (0.144)
<i>Panel B: Outcome = Depressed</i>			
All ($N = 8,821$)	-0.016* (0.010)	-0.016* (0.010)	-0.016 (0.010)
Male [†] ($N = 4,282$)	-0.008 (0.016)	-0.008 (0.015)	-0.008 (0.018)
Female ($N = 4,539$)	-0.029** (0.015)	-0.029** (0.014)	-0.029** (0.012)

Notes: All regressions use the single-instrument reduced-form specification (young $\times$ nin) with full controls. Pooled IFLS-4 and IFLS-5 with wave fixed effects. The coefficient is identical across clustering methods; only standard errors differ. Standard errors in parentheses. Column (1): heteroskedasticity-robust. Column (2): birth-district clusters (baseline). Column (3): province-level clusters based on current residence province; the corresponding province counts are 22 for All, 21 for Male, and 19 for Female after singleton adjustments. The female CES-D result remains statistically significant under all three inference schemes. Significance levels are based on unadjusted p -values. [†]Male pre-trends test is significant (see Table 3 notes for details). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A18: Robustness: Baseline Without vs. With Current Residence Province Fixed Effects

	(1)	(2)
	Baseline: no current	+ Current
	province FE	province FE
<i>Panel A: Outcome = CES-D 10</i>		
All ($N = 8,821$)	−0.273***	−0.316***
	(0.113)	(0.110)
Male [†] ($N = 4,282$)	−0.160	−0.174
	(0.185)	(0.193)
Female ($N = 4,539$)	−0.510***	−0.552***
	(0.155)	(0.159)
<i>Panel B: Outcome = Depressed</i>		
All ($N = 8,821$)	−0.016*	−0.017**
	(0.010)	(0.010)
Male [†] ($N = 4,282$)	−0.008	−0.007
	(0.015)	(0.015)
Female ($N = 4,539$)	−0.029**	−0.029**
	(0.014)	(0.014)

Notes: All regressions use the reduced-form specification ($\text{young} \times \text{nin}$) with full controls. Pooled IFLS-4 and IFLS-5 with wave fixed effects. Column (1) is the baseline specification used throughout the paper and excludes current residence province fixed effects, which could be post-treatment if INPRES affected migration patterns. Column (2) adds current residence province FE in addition to the baseline controls. Both columns include district FE, birth-year FE, baseline district trend controls, and individual demographics. Standard errors clustered at the birth-district level in parentheses. [†]Male pre-trends test is significant (see Table 3 notes for details). Significance levels are based on raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A19: Repeated-Observation and IFLS-5 Robustness

Sample	Outcome	N	IDs	Coef.	SE	Raw p	BH p
<i>Panel A: Baseline pooled specification</i>							
All	CES-D 10	8,821	5,144	−0.273**	0.113	0.016	0.074
All	Depressed	8,821	5,144	−0.016	0.010	0.114	0.209
Male	CES-D 10	4,282	2,560	−0.160	0.185	0.390	0.540
Male	Depressed	4,282	2,560	−0.008	0.015	0.586	0.703
Female	CES-D 10	4,539	2,584	−0.510***	0.155	0.001	0.021
Female	Depressed	4,539	2,584	−0.029**	0.014	0.033	0.107
<i>Panel B: Latest observation per person only</i>							
All	CES-D 10	5,144	5,144	−0.335*	0.173	0.054	0.121
All	Depressed	5,144	5,144	−0.021	0.014	0.142	0.233
Male	CES-D 10	2,560	2,560	−0.147	0.306	0.631	0.710
Male	Depressed	2,560	2,560	−0.003	0.023	0.883	0.883
Female	CES-D 10	2,584	2,584	−0.606***	0.223	0.007	0.051
Female	Depressed	2,584	2,584	−0.043**	0.021	0.044	0.113
<i>Panel C: IFLS-5 only</i>							
All	CES-D 10	4,215	4,215	−0.421**	0.199	0.036	0.107
All	Depressed	4,215	4,215	−0.022	0.017	0.200	0.299
Male	CES-D 10	2,031	2,031	−0.262	0.361	0.470	0.604
Male	Depressed	2,031	2,031	−0.008	0.027	0.762	0.762
Female	CES-D 10	2,184	2,184	−0.642***	0.241	0.008	0.051
Female	Depressed	2,184	2,184	−0.036	0.023	0.116	0.209

Notes: All rows report the coefficient on $\text{young}_i \times \text{nin}_{d(i)}$ from the preferred reduced-form specification with district FE, birth-year FE, baseline district trend controls, and individual demographics, excluding current residence province fixed effects to match the paper's baseline specification. Panel A reproduces the main pooled result. Panel B keeps only the latest observation for each person. Panel C estimates the same specification only in IFLS-5. Standard errors are clustered at the birth-district level. BH p -values apply Benjamini-Hochberg correction across all 18 estimates reported in the table. Significance stars are based on raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A20: Oster-Style Coefficient Stability for CES-D 10

Sample	Sparse coef.	Sparse R^2	Full coef.	Full R^2	$R_{\max}$	$\beta^*(\delta = 1)$	δ to zero
<i>Panel A: Outcome = CES-D 10</i>							
All	-0.246	0.119	-0.273	0.124	0.161	-0.476	-1.346
Male	-0.094	0.129	-0.160	0.137	0.178	-0.473	-0.510
Female	-0.482	0.149	-0.510	0.157	0.204	-0.672	-3.151

Notes: CES-D 10 only. Rows correspond to the All, Male, and Female pooled specifications from Table A16; $N = 8,821, 4,282$, and $4,539$, with 209, 195, and 187 birth-district clusters, respectively. Male estimates are reported for completeness but should be interpreted cautiously because the male pre-trends test fails. “Sparse” includes district FE, birth-year FE, and wave FE. “Full” adds baseline district trend controls and individual demographics, excluding current residence province fixed effects to match the paper’s baseline specification. $R_{\max}$ is set to $1.3 \times R_{\text{full}}^2$, capped at 1. $\beta^*(\delta = 1)$ is the Oster-adjusted coefficient under proportional selection on observables and unobservables. “ δ to zero” is the proportional-selection value required to explain the full-specification estimate away. Negative values indicate that omitted variables would need to move the estimate in the opposite direction from the observables to drive the coefficient to zero.

Table A21: Alternative Depression Cutoffs: Reduced-Form Estimates

Outcome	All	Male [†]	Female
CES-D 10 (continuous)	−0.273** (0.113)	−0.160 (0.185)	−0.510*** (0.155)
CES-D ≥ 8	−0.023* (0.012)	−0.010 (0.018)	−0.037** (0.015)
CES-D ≥ 10 (standard)	−0.016 (0.010)	−0.008 (0.015)	−0.029** (0.014)
CES-D ≥ 12	−0.008 (0.009)	−0.007 (0.011)	−0.016 (0.012)
CES-D ≥ 16	−0.010* (0.005)	−0.008 (0.006)	−0.015* (0.009)
<i>N</i>	8,821	4,282	4,539

Notes: All regressions use the single-instrument reduced-form specification ($\text{young} \times \text{nin}$) with full controls: district FE, birth-year FE, wave FE, baseline district trend controls, and individual demographics (birth month, religion [Muslim], ethnicity [Javanese], language [Indonesian speaker], parental education [mother and father]). Pooled IFLS-4 and IFLS-5 with wave fixed effects. Standard errors clustered at the birth-district level (209 clusters for All, 195 for Male, 187 for Female). The continuous CES-D 10 measure captures effects across the full distribution, while binary cutoffs test for shifts at specific clinical thresholds. The standard clinical cutoff for the CES-D 10 is ≥ 10 . INPRES cohorts (born 1958–62, 1968–72), $N = 8,821$. Significance levels are based on unadjusted p -values. Table 3 reports BH-adjusted stars for the six headline reduced-form outcomes; this appendix table does not. [†]Male pre-trends test is significant (see Table 3 notes for details). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A22: Leave-One-Province-Out Sensitivity: Reduced-Form CES-D 10 Estimates

Province excluded	All	Male [†]	Female
None (baseline)	−0.273*** (0.113)	−0.160 (0.185)	−0.510*** (0.155)
North Sumatra	−0.186 (0.127)	−0.095 (0.170)	−0.516*** (0.169)
West Sumatra	−0.221* (0.113)	−0.183 (0.190)	−0.452*** (0.153)
South Sumatra	−0.262** (0.114)	−0.210 (0.192)	−0.501*** (0.154)
Lampung	−0.268** (0.112)	−0.198 (0.193)	−0.482*** (0.151)
West Java	−0.324*** (0.117)	−0.297 (0.199)	−0.552*** (0.167)
Central Java	−0.311*** (0.115)	−0.200 (0.199)	−0.592*** (0.166)
DI Yogyakarta	−0.258** (0.124)	−0.206 (0.195)	−0.524*** (0.163)
East Java	−0.326*** (0.123)	−0.310 (0.203)	−0.456*** (0.163)
South Kalimantan	−0.278** (0.119)	−0.219 (0.192)	−0.511*** (0.164)
South Sulawesi	−0.247** (0.117)	−0.142 (0.205)	−0.484*** (0.166)

Notes: Each row excludes the named province and re-estimates the reduced-form specification (young $\times$ nin) with full controls on CES-D 10. N varies by excluded province and sample, starting from baseline totals of 8,821 (All), 4,282 (Male), and 4,539 (Female). Full controls: district FE, birth-year FE, wave FE, baseline district trend controls, and individual demographics. Pooled IFLS-4 and IFLS-5 with wave fixed effects. Standard errors clustered at the birth-district level; cluster counts vary by excluded province. Provinces not shown have too few INPRES-cohort observations to estimate the full specification; all other IFLS provinces are included. Significance levels are based on unadjusted p -values. [†]Male pre-trends test is significant (see Table 3 notes for details). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A23: Wild Cluster Bootstrap Inference for Main Reduced-Form Results

Sample	CES-D 10 Score			Depressed (≥ 10)		
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Male [†]	Female	All	Male [†]	Female
$\hat{\beta}$	-0.273** (0.113)	-0.160 (0.185)	-0.510*** (0.155)	-0.016 (0.010)	-0.008 (0.015)	-0.029** (0.014)
Analytical p -value	0.016	0.390	0.001	0.114	0.586	0.033
WCB p -value	0.049	0.397	0.000	0.120	0.589	0.023
N	8,821	4,282	4,539	8,821	4,282	4,539
Clusters	209	195	187	209	195	187

Notes: Each cell reports the reduced-form coefficient on $\text{young}_i \times \text{nin}_{d(i)}$ from Eq. (1). Pooled IFLS-4 and IFLS-5 with wave fixed effects. INPRES cohorts (born 1958–62, 1968–72). Controls: district fixed effects, birth-year fixed effects, wave fixed effects, baseline district trends (birth year $\times$ 1971 water/sanitation, enrollment, population), and individual demographics (birth month, religion [Muslim], ethnicity [Javanese], language [Indonesian speaker], parental education [mother and father]). Standard errors clustered at the birth-district level in parentheses. “Analytical p -value” is the standard cluster-robust p -value. “WCB p -value” is the wild cluster bootstrap p -value using the Webb 6-point distribution with 9,999 replications (`boottest` in Stata). [†]Male pre-trends test is significant (see Table 3 notes for details). Stars denote significance based on WCB p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A24: Sample Selection: Analysis Sample vs. Excluded INPRES-Cohort Observations

	All		Male		Female	
	(1)	(2)	(3)	(4)	(5)	(6)
	In	Out	In	Out	In	Out
<i>Means of key variables</i>						
Years of education	7.64	9.85	8.30	10.70	7.02	8.99
CES-D 10 score	4.84	4.99	4.65	4.79	5.02	5.18
Male	0.49	0.48	—	—	—	—
Muslim	0.89	0.89	0.90	0.87	0.89	0.92
Javanese ethnicity	0.47	0.20	0.48	0.23	0.46	0.17
INPRES intensity ($\text{nin}_{d(i)}$)	2.10	1.67	2.09	1.70	2.11	1.65
Young cohort	0.59	0.61	0.60	0.62	0.57	0.60
<i>N</i>	8,821	1,283	4,282	614	4,539	669

Notes: “In” = analysis sample; “Out” = INPRES-cohort observations excluded because of missing birth-district linkage or missing baseline/control variables. Excluded observations differ from the analysis sample mainly in geographic composition: they are less likely to be Javanese and come from lower-intensity districts on average, while mean CES-D scores are similar. This table is descriptive and reports means only.

Table A25: Complete-Case Inclusion Diagnostics

Sample	Inclusion on young $\times$ nin
<i>All</i> ($N = 9,542$)	
Analysis-sample inclusion	0.0010 (0.0010)
<i>Male</i> ($N = 4,644$)	
Analysis-sample inclusion	0.0009 (0.0010)
<i>Female</i> ($N = 4,898$)	
Analysis-sample inclusion	0.0008 (0.0014)

Notes: Each entry reports the coefficient from a linear-probability model for inclusion in the final complete-case analysis sample, estimated in the exposure-observed Duflo-cohort sample before complete-case restriction. Sample sizes are reported in the panel headers. The final paper sample remains $N = 8,821$ and $5,144$ unique individuals. Standard errors clustered at the birth-district level are in parentheses.

Table A26: Unweighted vs. Inverse-Probability-Weighted Reduced-Form Estimates

Outcome	Unweighted	IPW
<i>All (N = 8,821; IDs = 5,144)</i>		
CES-D 10	−0.2733** (0.1129)	−0.2733** (0.1131)
Depressed	−0.0161 (0.0101)	−0.0158 (0.0101)
<i>Male (N = 4,282; IDs = 2,560)</i>		
CES-D 10	−0.1597 (0.1854)	−0.1693 (0.1851)
Depressed	−0.0082 (0.0150)	−0.0086 (0.0149)
<i>Female (N = 4,539; IDs = 2,584)</i>		
CES-D 10	−0.5099*** (0.1547)	−0.5139*** (0.1555)
Depressed	−0.0294** (0.0136)	−0.0296** (0.0137)

Notes: All regressions use the preferred reduced-form specification and are reported on the final analysis sample used in the paper ($N = 8,821$, 5,144 unique individuals). The weighted column uses inverse predicted inclusion probabilities estimated from the exposure-observed sample but applied to the same final analysis sample. Standard errors clustered at the birth-district level are in parentheses. Significance stars are based on raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A27: Selection Balance: Observables by INPRES Construction Intensity

	Below-median nin	Above-median nin	Difference	SE	<i>p</i> -value
Years of education	8.17	7.11	−1.101***	(0.323)	0.001
Male	0.493	0.477	−0.015	(0.013)	0.250
Muslim	0.937	0.849	−0.087*	(0.045)	0.053
Javanese ethnicity	0.578	0.358	−0.220***	(0.082)	0.008
Mother educated	0.708	0.662	−0.045**	(0.018)	0.016
Father educated	0.777	0.742	−0.035**	(0.017)	0.042
CES-D 10	4.64	5.04	0.354*	(0.182)	0.054
Depressed (≥ 10)	0.116	0.139	0.020*	(0.011)	0.073
<i>N</i>	4,460	4,361			

Notes: Means of observables and outcomes by above/below median INPRES construction intensity (nin) within the complete-case analysis sample used in the main regressions (Table A2, $N = 8,821$; 209 birth-district clusters). The two N columns therefore sum to the full analysis sample. Differences are regression-adjusted with birth-year FE and wave FE, with standard errors clustered at the birth-district level. Above-median districts have lower education and parental education, reflecting the program’s redistributive allocation rule; these compositional differences are absorbed by district fixed effects in the main analysis. The depression outcomes are somewhat higher in above-median districts descriptively, but those differences are only marginally significant after adjusting for cohort composition. Pooled IFLS-4 and IFLS-5. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A28: IFLS-1-Linked Selection Diagnostic Sample Construction

Step	<i>N</i>
<i>Panel A: Selection-diagnostic flow</i>	
IFLS-1 baseline members	33,081
IFLS-1 Duflo cohorts	4,546
Duflo cohorts with birth district observed	4,286
Duflo cohorts with INPRES intensity observed	4,270
Final IFLS-1-linked selection sample	3,913
Linked to live IFLS-4/IFLS-5 main-sample frame	3,332
<i>Panel B: Exclusions from 4,270 to 3,913</i>	
Missing 1971 water/sanitation control (<i>wsppc</i>)	279
Missing 1971 population control (<i>pop71</i>)	57
Missing 1971 enrollment control (<i>en71</i>)	21

Notes: This table summarizes the baseline universe used for the IFLS-1-linked survival and attrition diagnostic. The diagnostic sample is smaller than the paper's IFLS-4/IFLS-5 outcome sample because later death and non-observation must be defined relative to a common IFLS-1 baseline panel. The reduction from 4,270 to 3,913 is a baseline covariate-availability restriction, driven by missing district-level 1971 controls rather than later follow-up loss. The final row in Panel A indicates how many members of the IFLS-1-linked diagnostic sample can be linked to the live IFLS-4/IFLS-5 main-sample frame used for the depression analysis.

Table A29: IFLS-1-IFLS-5 Selection Diagnostics: By-Wave Mortality and Attrition

Wave	All	Male	Female
<i>Panel A: Outcome = Dead by follow-up wave</i>			
1997 ($N = 3,913/1,844/2,069$)	−0.0063 (0.0042)	−0.0080 (0.0075)	−0.0030 (0.0041)
2000 ($N = 3,913/1,844/2,069$)	−0.0070 (0.0050)	−0.0044 (0.0083)	−0.0075 (0.0061)
2007 ($N = 3,913/1,844/2,069$)	−0.0037 (0.0091)	−0.0128 (0.0156)	0.0028 (0.0091)
2014 ($N = 3,913/1,844/2,069$)	−0.0044 (0.0103)	−0.0041 (0.0153)	−0.0077 (0.0135)
<i>Panel B: Outcome = Alive but not observed by follow-up wave</i>			
1997 ($N = 3,887/1,828/2,059$)	−0.0072 (0.0144)	−0.0248 (0.0266)	0.0151 (0.0172)
2000 ($N = 3,859/1,815/2,044$)	0.0050 (0.0133)	−0.0149 (0.0243)	0.0231 (0.0167)
2007 ($N = 3,773/1,769/2,004$)	0.0012 (0.0121)	−0.0121 (0.0247)	0.0153 (0.0161)
2014 ($N = 3,646/1,699/1,947$)	0.0174 (0.0127)	0.0079 (0.0235)	0.0312* (0.0174)

Notes: Each cell reports the coefficient on $\text{young}_i \times \text{nin}_{d(i)}$ from a by-wave linear-probability model estimated on the IFLS-1-linked baseline panel. Panel A uses cumulative death-by-wave as the outcome. Panel B uses alive-but-not-observed attrition, conditioning on not being dead by that wave. All columns include birth-district FE, birth-year FE, birth-month FE, and the paper's baseline district trend controls (birth year interacted with 1971 water/sanitation, enrollment, and population). The pooled column additionally includes a female indicator; sex-specific columns omit it. Standard errors clustered at the birth-district level are in parentheses (193 clusters for All, 179 for Male, 178 for Female). Significance stars are based on raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A30: Wave-Specific Reduced-Form Results (Preferred Specification)

	(1)	(2)	(3)
	Combined	IFLS-4 Only	IFLS-5 Only
	(IFLS-4 + IFLS-5)	(2007)	(2014)
<i>Panel A: All ($N = 8,821/4,606/4,215$)</i>			
CES-D 10	-0.273**	-0.156	-0.422**
	(0.113)	(0.106)	(0.199)
Depressed	-0.016	-0.011	-0.022
	(0.010)	(0.010)	(0.017)
<i>Panel B: Male[†] ($N = 4,282/2,251/2,031$)</i>			
CES-D 10	-0.160	-0.077	-0.262
	(0.185)	(0.165)	(0.361)
Depressed	-0.008	-0.007	-0.008
	(0.015)	(0.016)	(0.027)
<i>Panel C: Female ($N = 4,539/2,355/2,184$)</i>			
CES-D 10	-0.510***	-0.375**	-0.642***
	(0.155)	(0.182)	(0.241)
Depressed	-0.029**	-0.021	-0.036
	(0.014)	(0.014)	(0.023)

Notes: Each cell reports the reduced-form coefficient on $\text{young} \times \text{nin}$ from the preferred specification (Eq. (1) with full controls). The “Combined” column pools IFLS-4 (2007) and IFLS-5 (2014) with wave fixed effects. “IFLS-4 Only” and “IFLS-5 Only” estimate the specification separately on each wave without wave fixed effects. All specifications include district fixed effects, birth-year fixed effects, baseline district trend controls (birth year $\times$ 1971 water/sanitation, enrollment, population), and individual demographics (birth month, religion, ethnicity, language, parental education), but exclude current residence province fixed effects to match the paper’s baseline specification. Standard errors clustered at the birth-district level in parentheses. INPRES cohorts (born 1958–62, 1968–72). [†]Male pre-trends test is significant (see Table 3 notes for details). Cluster counts: 209 (All), 195 (Male), 187 (Female) for the combined specification; single-wave estimates use the same birth districts but with fewer observations per cluster. Significance levels are based on raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A31: Wave-Specific First-Stage F-Statistics (Single Instrument: $\text{young} \times \text{nin}$)

Wave	Sample	First-stage F
<i>Combined (IFLS-4 + IFLS-5)</i>		
	All ($N = 8,821$)	1.20
	Male [†] ($N = 4,282$)	3.43
	Female ($N = 4,539$)	0.16
<i>IFLS-4 Only (2007)</i>		
	All ($N = 4,606$)	0.82
	Male [†] ($N = 2,251$)	2.68
	Female ($N = 2,355$)	0.05
<i>IFLS-5 Only (2014)</i>		
	All ($N = 4,215$)	0.92
	Male [†] ($N = 2,031$)	1.86
	Female ($N = 2,184$)	0.27

Notes: First-stage F -statistics from the regression of years of education on $\text{young}_i \times \text{nin}_{d(i)}$ (single instrument), by wave and sample. The “Combined” specification pools IFLS-4 (2007) and IFLS-5 (2014) with wave fixed effects. “IFLS-4 Only” and “IFLS-5 Only” estimate the specification separately on each wave without wave fixed effects. With a single instrument, these are the first-stage test statistics on $\text{young} \times \text{nin}$; Table A30 reports the corresponding reduced-form coefficients and p -values. All specifications use the full control set: district fixed effects, birth-year fixed effects, baseline district trend controls, and individual demographics, excluding current residence province fixed effects to match the paper’s baseline specification. Standard errors clustered at the birth-district level. INPRES cohorts (born 1958–62, 1968–72). [†]Male pre-trends test is significant (see Table 3 notes for details).

Table A32: Panel Fixed Effects: Education and Depression (Individual FE, Within-Person)

	(1)
	$\hat{\beta}_{\text{educ}}$
<i>Panel A: Outcome = CES-D 10</i>	
All	0.001 (0.031)
Male	0.001 (0.041)
Female	0.000 (0.039)
<i>Panel B: Outcome = Depressed</i>	
All	0.001 (0.002)
Male	0.003 (0.003)
Female	−0.002 (0.004)
<i>N</i> (observations: All / Male / Female)	8,821 / 4,282 / 4,539
Individuals (All / Male / Female)	5,144 / 2,560 / 2,584

Notes: Individual fixed effects (within-estimator) using the unbalanced person-wave panel within the INPRES cohort analysis sample (born 1958–62 or 1968–72) with complete controls. The coefficient $\hat{\beta}_{\text{educ}}$ is on years of education. Controls include wave fixed effects and a quadratic age polynomial. Standard errors clustered at the birth-district level (cluster counts: 209 All, 195 Male, 187 Female). The near-zero coefficients reflect limited within-person variation in education across waves: 71% of individuals report the same years of schooling in both waves. This motivates the cross-sectional reduced-form approach, where the INPRES instrument—being time-invariant—cannot be used in a fixed-effects framework. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A33: Formal Java vs. Non-Java Interaction Tests

Sample	Outcome	Java	Non-Java	Difference	Raw p	BH p
<i>All</i> ($N = 8,821$)						
	CES-D 10	0.206 (0.184)	-0.557 (0.148)	-0.763 (0.236)	0.001	0.009
	Depressed	0.021 (0.015)	-0.038 (0.013)	-0.059 (0.020)	0.004	0.012
<i>Male</i> [†] ($N = 4,282$)						
	CES-D 10	0.348 (0.347)	-0.527 (0.265)	-0.876 (0.436)	0.046	0.073
	Depressed	0.031 (0.022)	-0.031 (0.022)	-0.062 (0.031)	0.049	0.073
<i>Female</i> ($N = 4,539$)						
	CES-D 10	-0.092 (0.339)	-0.712 (0.215)	-0.620 (0.401)	0.124	0.124
	Depressed	0.009 (0.029)	-0.053 (0.016)	-0.062 (0.033)	0.061	0.073

Notes: Each row comes from a pooled IFLS-4 and IFLS-5 reduced-form regression that interacts $\text{young}_i \times \text{nin}_{d(i)}$ with a non-Java indicator and includes the lower-order geography interactions needed for a formal difference-in-coefficients test. “Difference” equals the non-Java coefficient minus the Java coefficient. All specifications include district FE, birth-year FE, wave FE, baseline district trend controls, and individual demographics, excluding current residence province fixed effects to match the paper’s baseline specification. Standard errors clustered at the birth-district level in parentheses. BH p -values apply Benjamini-Hochberg correction across the 6 reported interaction tests. [†]Male estimates do not pass pre-trends tests and are reported only for completeness.

Table A34: Wave-Specific Java vs. Non-Java Reduced-Form Estimates

Sample	Java	Non-Java	Difference	Raw p
<i>Panel A: IFLS-4, Outcome = CES-D 10</i>				
All ($N = 4,606$)	-0.2042 (0.2014)	-0.1855 (0.1382)	0.0187 (0.2442)	0.9390
Male ($N = 2,251$)	0.0313 (0.2865)	-0.2756 (0.2329)	-0.3068 (0.3693)	0.4070
Female ($N = 2,355$)	-0.6485* (0.3584)	-0.2502 (0.2274)	0.3983 (0.4245)	0.3493
<i>Panel B: IFLS-4, Outcome = Depressed</i>				
All ($N = 4,606$)	-0.0048 (0.0139)	-0.0081 (0.0114)	-0.0033 (0.0180)	0.8552
Male ($N = 2,251$)	0.0112 (0.0172)	-0.0071 (0.0200)	-0.0183 (0.0263)	0.4891
Female ($N = 2,355$)	-0.0277 (0.0220)	-0.0178 (0.0172)	0.0099 (0.0279)	0.7242
<i>Panel C: IFLS-5, Outcome = CES-D 10</i>				
All ($N = 4,215$)	0.7182** (0.3364)	-1.0025*** (0.2268)	-1.7207*** (0.4057)	0.0000
Male ($N = 2,031$)	0.5908 (0.5940)	-0.8559 (0.5275)	-1.4467* (0.7944)	0.0702
Female ($N = 2,184$)	0.6420 (0.5435)	-1.1684*** (0.3362)	-1.8105*** (0.6391)	0.0051
<i>Panel D: IFLS-5, Outcome = Depressed</i>				
All ($N = 4,215$)	0.0530* (0.0307)	-0.0716*** (0.0208)	-0.1246*** (0.0371)	0.0009
Male ($N = 2,031$)	0.0508 (0.0463)	-0.0578 (0.0428)	-0.1085* (0.0631)	0.0869
Female ($N = 2,184$)	0.0563 (0.0568)	-0.0857*** (0.0245)	-0.1420** (0.0618)	0.0229

Notes: All regressions are estimated separately by wave, within the Java and non-Java subsamples, using the preferred reduced-form specification on INPRES cohorts born 1958–62 and 1968–72. Controls include district FE, birth-year FE, baseline district trend controls, and individual demographics. Standard errors clustered at the birth-district level are in parentheses. Cluster counts are 100/106 (All), 100/93 (Male), and 94/90 (Female) in IFLS-4 for Java/Non-Java, and 98/100 (All), 98/87 (Male), and 92/87 (Female) in IFLS-5 for Java/Non-Java. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A35: IFLS-5 Java Leave-One-Province-Out Sensitivity

Excluded province (<i>N</i> all / male / female)	All	Male	Female
<i>Panel A: Outcome = CES-D 10</i>			
Baseline (2,427 / 1,177 / 1,250)	0.6237* (0.3407)	0.5347 (0.5497)	0.4527 (0.5266)
Exclude West Java (1,846 / 908 / 938)	0.2568 (0.3611)	0.5365 (0.6006)	−0.1623 (0.5544)
Exclude Central Java (1,642 / 796 / 846)	1.1102*** (0.3576)	1.3331** (0.6352)	0.6882 (0.5364)
Exclude DI Yogyakarta (2,123 / 1,024 / 1,099)	0.7069* (0.3903)	0.8577 (0.6306)	0.2387 (0.5691)
Exclude East Java (1,670 / 803 / 867)	0.6123 (0.5436)	−0.6726 (0.7717)	1.2886 (0.8290)
<i>Panel B: Outcome = Depressed</i>			
Baseline (2,427 / 1,177 / 1,250)	0.0495 (0.0307)	0.0474 (0.0435)	0.0388 (0.0568)
Exclude West Java (1,846 / 908 / 938)	0.0158 (0.0318)	0.0391 (0.0525)	−0.0104 (0.0580)
Exclude Central Java (1,642 / 796 / 846)	0.0904** (0.0357)	0.1255** (0.0496)	0.0398 (0.0714)
Exclude DI Yogyakarta (2,123 / 1,024 / 1,099)	0.0526 (0.0335)	0.0747 (0.0504)	0.0221 (0.0609)
Exclude East Java (1,670 / 803 / 867)	0.0558 (0.0467)	−0.0521 (0.0589)	0.1247 (0.0766)

Notes: Leave-one-province-out sensitivity restricted to IFLS-5 Java observations in the IN-PRES cohorts (born 1958–62 and 1968–72). Outcome rows labeled Depressed use Depressed = 1 if CES-D 10 $\geq$ 10. Each row drops one Java province (code in parentheses) and re-estimates the preferred reduced-form specification. All regressions include district FE, birth-year FE, baseline district trends, and individual demographics. Standard errors clustered at the birth-district level are in parentheses. Baseline cluster counts are 98 (All), 98 (Male), and 92 (Female). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A36: Reduced-Form: Java vs. Non-Java Provinces (IFLS-4 Only, 2007)

	(1)	(2)
	Java	Non-Java
<i>Panel A: All</i> ($N_{Java} = 2,702$; $N_{Non-Java} = 1,904$)		
CES-D 10	-0.220	-0.196
	(0.196)	(0.154)
Depressed	-0.010	-0.007
	(0.014)	(0.013)
<i>Panel B: Male</i> [†] ($N_{Java} = 1,334$; $N_{Non-Java} = 917$)		
CES-D 10	0.108	-0.188
	(0.296)	(0.226)
Depressed	0.014	-0.006
	(0.018)	(0.023)
<i>Panel C: Female</i> ($N_{Java} = 1,368$; $N_{Non-Java} = 987$)		
CES-D 10	-0.676*	-0.351
	(0.362)	(0.266)
Depressed	-0.041*	-0.018
	(0.023)	(0.021)

Notes: Each cell reports the coefficient on $\text{young} \times \text{nin}$ from the preferred reduced-form specification estimated separately within the Java and non-Java subsamples. IFLS-4 wave only, so wave fixed effects are omitted. Controls include district FE, birth-year FE, baseline district trend controls, and individual demographics. Standard errors clustered at the birth-district level in parentheses. Clusters: Java All = 100, Male = 100, Female = 94; Non-Java All = 106, Male = 93, Female = 90. [†]Male pre-trends test is significant (see Table 3 notes for details). These wave-specific splits are descriptive; Table A33 reports the formal pooled interaction tests. Significance levels are based on raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A37: Reduced-Form: Java vs. Non-Java Provinces (IFLS-5 Only, 2014)

	(1)	(2)
	Java	Non-Java
<i>Panel A: All</i> ($N_{Java} = 2,427$; $N_{Non-Java} = 1,788$)		
CES-D 10	0.624*	-0.927***
	(0.341)	(0.237)
Depressed	0.050	-0.066***
	(0.031)	(0.023)
<i>Panel B: Male</i> [†] ($N_{Java} = 1,177$; $N_{Non-Java} = 854$)		
CES-D 10	0.535	-0.731
	(0.550)	(0.553)
Depressed	0.047	-0.058
	(0.044)	(0.044)
<i>Panel C: Female</i> ($N_{Java} = 1,250$; $N_{Non-Java} = 934$)		
CES-D 10	0.453	-1.074***
	(0.527)	(0.305)
Depressed	0.039	-0.074***
	(0.057)	(0.026)

Notes: Each cell reports the coefficient on $\text{young} \times \text{nin}$ from the preferred reduced-form specification estimated separately within the Java and non-Java subsamples. IFLS-5 wave only, so wave fixed effects are omitted. Controls include district FE, birth-year FE, baseline district trend controls, and individual demographics. Standard errors clustered at the birth-district level in parentheses. Clusters: Java All = 98, Male = 98, Female = 92; Non-Java All = 100, Male = 87, Female = 87. The Java coefficients in this table therefore match the baseline row of Table A35, which re-estimates the same IFLS-5 Java-only regression before dropping individual provinces. [†]Male pre-trends test is significant (see Table 3 notes for details). These wave-specific splits are descriptive; Table A33 reports the formal pooled interaction tests. Significance levels are based on raw p -values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.